UNIT & DIMENSION

Passage - 1

The van-der Waals equation is

$$\left(P+\frac{a}{V^2}\right)(V-b)=RT,$$

where P is pressure, V is molar volume and T is the temperature of the given sample of gas. R is called molar gas constant, a and b are called van-der Wall constants.

1. The dimensional formula for b is same as that for

(A) P *(B) V (C) PV^2 (D) RT

2. The dimensional formula for a is same as that for

(A) V^2 (B) P *(C) PV^2 (D) RT

3. Which of the following does not possess the same dimensional formula as that for RT

(A) PV (B) Pb *(C) a/V^2 (D) ab/V^2

VECTOR

Passage-2

Two vectors $\vec{a}$ and $\vec{b}$ are varying with time as $\vec{a}=3t\hat{i}+4t^2\hat{j}$ and $\vec{b}=(6t+3)\hat{i}+(7\sin t)\hat{j}$.

4. Find the magnitude of the rate of change of $\vec{a}\times\vec{b}$ at $t=\pi/2$ sec.

*(A) $3[7-6\pi^2-4\pi]$ (B) $3[6\pi^2+4\pi-7]$

(C) $3[6\pi^2-4\pi+7]$ (D) $3[6\pi^2-4\pi-7]$

5. Find the rate of change of $\vec{a}.\vec{b}$ at $t=0$

(A) 36 *(B) 9 (C) 28 (D) 45

6. Find the rate of change of $\vec{a}.(\vec{a}\times\vec{b})$ at $t=0$

(A) 72 (B) 36 (C) 48 *(D) zero

KINEMATICS

Passage- 3

A particle of mass 2 kg starts to move at position x = 0 and time t = 0 under the action of force F = (10 + 4x) N along x-axis on a frictionless track and moved to a distance 5m.

7. The velocity of the particle in m/s is

(A) 5 *(B) 10 (C) 20 (D) 30

8. The force at that moment in N is

(A) 20 (B) 5 *(C) 30 (D) 40

9. The instantaneous power is

(A) 200 W *(B) 300 W (C) 400 W (D) 500 W

Passage -4

A solid body rotates about a stationary axis according to the law $\theta = kt - bt^3$, where k = 6.0 rad/s and b = 2.0 rad/s^2.

10. The mean value of the angular velocity averaged over the time interval between t = 0 and the complete stop is

(A) 2 rad/s *(B) 4 rad/s (C) 3 rad/s (D) 12 rad/s

11. The mean value of the angular acceleration averaged over the time interval between t = 0 and the complete stop is
*(A) 6 rad/s^2 (B) 24 rad/s^2 (C) 18 rad/s^2 (D) 12 rad/s^2

12. The angular acceleration at the moment when the body stops is
(A) 18 rad/s^2 (B) 24 rad/s^2 (C) 6 rad/s^2 *(D) 12 rad/s^2

Passage-5

A boat of mass M sails with velocity $v_0\hat{i}$. At t = 0, its engine stops, and at the same time a ball of mass m(m<<M) is thrown from the boat with initial velocity $u\hat{j}$. Here x-direction is horizontal and y is vertically up. The water exerts a friction drag force proportional to the boat's velocity (v), so that $\vec{F} = kv\hat{i}$, where K is positive constant. The acceleration of the boat can be written as :

$$\vec{a} = -\frac{kv}{M}\hat{i} \text{ or } \frac{d\vec{v}}{dt} = -\frac{kv}{M}\hat{i} \text{ integrating we get :}$$

$\vec{v} = \left(Ce^{-Kt/M}\right)\hat{i}$ where C is a constant emerging due to integrating. Now assume that ball is observed from the reference frame of boat (X'Y'). The origin of this reference frame coincided with the static frame (XY) at t = 0. The acceleration of the ball in the reference frame of boat can be written as $\vec{a'} = a_{x'}\hat{i} + a_{y'}\hat{j}$ and in the static frame it can be written as $\vec{a} = a_x\hat{i} + a_y\hat{j}$.

13. The value of constant C is :
(A) $v_0/2$ *(B) v_0 (C) $v_0(M/m)$ (D) $v_0(m/M)$

14. The acceleration of boat as a function of time is :
(A) $\frac{Kv_0}{M}e^{-(Kt/M)}\hat{i}$ (B) $\frac{Kv_0}{M}e^{-(Kt/M)}\left(-\hat{i}\right)$ *(C) $-\frac{Kv_0}{M}e^{-(Kt/M)}\hat{i}$ (D) $-\frac{Kv_0}{M}e^{(Kt/M)}\hat{i}$

15. Values of a_x and $a_{x'}$ respectively are :
*(A) $a_x = 0;\ a_{x'} = \frac{Kv_0}{M}e^{-(Kt/M)}$ (B) $a_x = 0;\ a_{x'} = -\frac{Kv_0}{M}e^{-(Kt/M)}$
(C) $a_x = a_{x'} = -\frac{Kv_0}{M}e^{-(Kt/M)}$ (D) $a_x = 0;\ a_{x'} = -\frac{Kv_0}{M}e^{(Kt/M)}$

Passage-6

A particle starting from rest has a constant acceleration of 4 m/s^2 for 4 seconds. It then retards uniformly for next 8 seconds and comes to rest.

16. Average acceleration during the motion of the particle is
(A) 4 m/s^2 *(B) zero (C) 8 m/s^2 (D) –4 m/s^2

17. Average speed during the motion of the particle is
*(A) 8 m/s (B) zero (C) 4 m/s (D) 16 m/s

18. Average velocity during the motion of the particle is
(A) zero *(B) 8 m/s (C) 2 m/s (D) 4 m/s

Passage-7

A radius vector of a point A relative to the origin varies with time as $\vec{r} = at\hat{i} - bt^2\hat{j}$, where a and b are position coordinates and $\hat{i}$ and $\hat{j}$ are the unit vectors of the x and y axes. Find

19. The equation of the point's trajectory i.e., relation between y and x position of point.
(A) $y = x^2\frac{b}{a^2}$ (B) $y = -x\frac{b}{a^2}$ *(C) $y = -x^2\frac{b}{a^2}$ (D) $y = -x^2\frac{b}{a}$

20. The time dependence of the velocity $\vec{v}$ is

(A) $\vec{v} = -a\hat{i} - 2bt\hat{j}$ (B) $\vec{v} = a\hat{i} - 2b\hat{j}$ *(C) $\vec{v} = a\hat{i} - 2bt\hat{j}$ (D) $\vec{v} = a\hat{i} + 2bt\hat{j}$

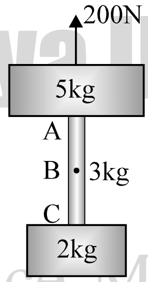
21. The time dependence of the acceleration $\vec{\omega}$ is

(A) $\vec{\omega} = 2a\hat{j}$ (B) $\vec{\omega} = 2b^2\hat{j}$ (C) $\vec{\omega} = 2b\hat{j}$ *(D) $\vec{\omega} = -2b\hat{j}$

NEWTON'S LAWS OF MOTION

Passage-8

With reference to the figure shown below, answer the following questions.



22. The value of tension at A is

*(A) 100 N (B) 150 N (C) 200 N (D) 50 N

23. The value of tension at B is

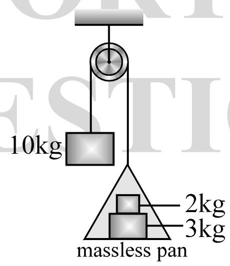
(A) 60 N (B) 40 N *(C) 70 N (D) 100 N

24. The value of tension at C is

(A) 10 N (B) 30 N (C) 20 N *(D) 40 N

Passage-9

A pulley mass system is shown below. The pulley and the strings are massless, then



25. Force applied by 3 kg block on 2 kg block is

*(A) 8g/3 N (B) 5g/3 N (C) 20g/3 N (D) 10g/3 N

26. Force applied by 3 kg block on pan is

(A) 8g/3 N (B) 5g/3 N *(C) 20g/3 N (D) 10g/3 N

27. Tension in the string is

(A) 8g/3 N (B) 5g/3 N *(C) 20g/3 N (D) 10g/3 N

Passage-10

Experiment 1

The student pushes horizontally (rightward) on the crate of mass 100 kg and gradually increases the strength of this pushing force. The crate does not begin to move until the push force reaches 400 N.

Experiment 2

The student applies a constant horizontal (rightward) push force for 1.0 s and measure how far the crate moves during that time interval. In each trial the crate starts at rest and the student stops pushing after the 1.0 s interval. The following table summarizes the results.

Trial	Push force (N)	Distance (m)
1	500	1.5
2	600	2
3	700	2.5

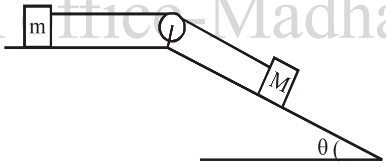
28. The coefficient of static friction between the crate and the floor is approximately
(A) 0.25 (B) 2.5 *(C) 0.40 (D) 4.0

29. In experiment, when the rightward push force was 50 N, the crate didn't move. Why didn't it move ?
(A) The push force was weaker than frictional force on the crate.
(B) The push force has the same strength as the gravitational force on the crate.
*(C) The push force had the same strength as the frictional force on the crate.
(D) The push force was stronger than the frictional force on the crate.

30. The coefficient of kinetic friction between the crate and the floor is approximately
(A) 0.30 *(B) 0.20 (C) 0.40 (D) 0.50

Passage-11

Consider the system shown in figure. Friction coefficient at both the contacts is μ



31. Maximum value of M/m for which the system remains at rest
(A) $\dfrac{\mu}{\cos\theta+\mu\sin\theta}$ (B) $\dfrac{\mu}{\cos\theta-\mu\sin\theta}$ (C) $\dfrac{\mu}{\sin\theta+\mu\cos\theta}$ *(D) $\dfrac{\mu}{\sin\theta-\mu\cos\theta}$

32. When $\tan\theta < \mu$
*(A) The system will not slide for any value of M/m.
(B) The mass M will move upwards
(C) The mass M will lose contact with the inclined plane.
(D) The two bodies will move at constant velocity

33. If the value of M/m is twice that derived in question 14. Find the acceleration of mass M.
(A) $[Mg\sin\theta+\mu g(m+M\cos\theta)]/m+M$ (B) $[Mg\cos\theta+\mu g(m+M\sin\theta)]/m+M$
*(C) $[Mg\sin\theta-\mu g(m+M\cos\theta)]/m+M$ (D) $[Mg\cos\theta-\mu g(m+M\sin\theta)]/m+M$

Passage-12

A 2 kg block is placed over a 4 kg block and both are placed on a smooth horizontal surface. The coefficient of friction between the blocks is 0.20.

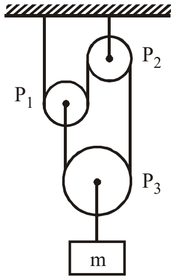
34. The acceleration of the upper block if a horizontal force of 12 N is applied to the upper block is
(A) 2 m/s^2 (B) 1 m/s^2 *(C) 4 m/s^2 (D) 5 m/s^2

35. The acceleration of the lower block if a horizontal force of 12 N is applied to the upper block is
(A) 2 m/s^2 *(B) 1 m/s^2 (C) 4 m/s^2 (D) 5 m/s^2

36. The acceleration of the upper block if a horizontal force of 12 N is applied to the lower block is
*(A) 2 m/s^2 (B) 1 m/s^2 (C) 4 m/s^2 (D) 5 m/s^2

Passage-13

Three smooth pulleys are connected by a light string. A mass m is attached to the pulley P_3. Assume the strings between two pulleys and between the pulley and the support to be vertical.



37. If all the pulleys are massless then the tension in the string over the pulleys is
(A) mg *(B) T = 0 (C) 2mg (D) mg/2

38. The relation between the magnitudes of acceleration of pulley P_1 (a_{P_1}) and that of pulley P_3(a_{P_3}) is
(A) $a_{P_1} = a_{P_3}$ (B) $2a_{P_1} = a_{P_3}$ *(C) $a_{P_1} = 2a_{P_3}$ (D) $a_{P_1} = 3a_{P_3}$

39. The tension in the string over the pulleys will be non zero when
(A) mass of pulley P_1 is zero (B) mass of pulley P_2 is zero
(C) mass of pulley P_2 is non zero *(D) mass of pulley P_1 is non zero

Passage-14

A pendulum bob of mass 50g is suspended from the ceiling of an elevator. (g = 10 m/s^2)

40. The tension in the string if the elevator goes up with acceleration 1.2 m/s^2 is
(A) 0.60 N *(B) 0.56 N (C) 0.75 N (D) 0.45 N

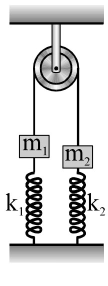
41. The tension in the string if the elevator goes up with uniform velocity is
*(A) 0.5 N (B) 0.43 N (C) 0.52 N (D) 0.75 N

42. The tension in the string if the elevator goes down with decceleration 1.2 m/s^2 is
(A) 0.49 N (B) 0.43 N *(C) 0.56 N (D) 0.75 N

WORK POWER & ENERGY

Passage- 15

The system shown in the figure is in equilibrium. Masses m_1 and m_2 are 2 kg and 8 kg respectively. Spring constants k_1 and k_2 are 50 N/m and 70 N/m respectively. The compression in second spring is 0.5 m. Both springs have the same natural length. (g = 10 m/s^2)



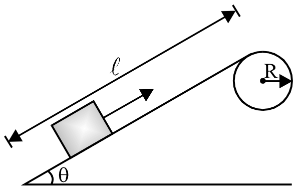
43. The compression in first spring
(A) 1.3 m (*B) –0.5 m (C) 0.5 m (D) 0.9 m

44. Find the acceleration of the mass m_1 when spring k_1 is cut
(A) zero (*B) 2.5 m/s^2 (C) 3 m/s^2 (D) none

45. Find the acceleration of the block m_1 when the string is cut
(A) 20 m/s^2 (*B) 22.5 m/s^2 (C) 25 m/s^2 (C) 25.5 m/s^2

Passage -16

Figure shows a smooth track which consists of a straight inclined part of length ℓ joining smoothly with the circular part. A particle of mass m is projected up the incline from its bottom then



46. The minimum projection speed v_0 for which the particle reaches the top of the track

(A) $\sqrt{2g(\ell\sin\theta + R\cos\theta)}$ (B) $\sqrt{2gR(1-\cos\theta)}$

(C) $\sqrt{2g\ell\sin\theta}$ (*D) $\sqrt{2g[R(1-\cos\theta)+\ell\sin\theta]}$

47. If the particle is projected with a speed twice the minimum speed (for which the particle reaches the top of the track) and that block does not lose contact with the track before reaching its top, the force acting on it when it reaches the top is

(*A) $6mg\left(1-\cos\theta+\frac{\ell}{R}\sin\theta\right)$ (B) 6mg (1-cosθ)

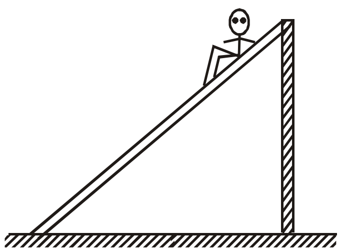
(C) 6 mg (D) 6 mgcosθ

48. If the projection speed is slightly greater than the speed at which the particle reaches the top of the track then, the block will lose contact with the track, when radius through the particle makes an angle .... with the vertical

(A) $\cos^{-1}\left(\frac{1}{3}\right)$ (*B) $\cos^{-1}\left(\frac{2}{3}\right)$ (C) $\cos^{-1}\left(\frac{3}{5}\right)$ (D) $\cos^{-1}\left(\frac{4}{5}\right)$

Passage-17

In a children's park, there is a slide which has a total length of 10 m and a height of 8.0m as shown in figure. A vertical ladder is provided to reach the top. A body weighing 200 N climbs up the ladder to the top of the slide and slides down the ground. The average friction offered by the slide is three tenth of his weight.



49. The work done by the ladder on the boy as he goes up is

(A) 600 J (B) 1600 J *(C) zero (D) 800 J

50. The work done by the slide on the boy as he comes down is

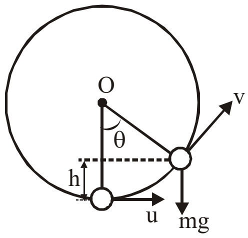
(A) 1600 J *(B) –600 J (C) zero (D) 500 J

51. The work done by forces inside the body of the boy is

(A) zero (B) –600 J (C) 800 J *(D) 1600 J

Passage-18

A particle of mass M attached to an inextensible string is moving in a vertical circle of radius R about fixed point O. It is imparted a velocity u in horizontal direction at the lowest position as shown in figure. ($g = m/s^2$)



52. If R = 2m, M = 2kg and u = 12 m/s then, value of tension at lowest position is
(A) 120 N *(B) 165 N (C) 264 N (D) zero

53. Tension at highest point of its trajectory in above question will be
(A) 100 N *(B) 44 N (C) 144 N (D) 264 N

54. If M = 2kg, R = 2m and u = 10 m/s, the velocity of particle when $\theta = 60^o$
(A) $2\sqrt{5}$ m/s *(B) $4\sqrt{5}$ m/s (C) $5\sqrt{2}$ m/s (D) 5 m/s

Passage-19

One end of massless inextensible string of length l is fixed and the other end is tied to a small ball of mass m. The ball is performing a circular motion in vertical plane. At the lowest position, the speed of the ball is $\sqrt{20gl}$. Neglect any other forces on the ball except tension force and gravitational force. Acceleration due to gravity is g.

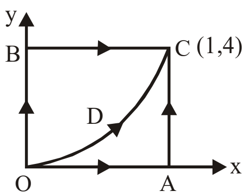
55. Motion of ball is in nature of
*(A) circular motion with variable speed
(B) circular motion with constant speed
(C) circular motion with constant angular acceleration about centre of the circle.
(D) None of these

56. At the highest position of the ball, tangential acceleration is
(A) g *(B) 0 (C) 5g (D) 16 g

57. During circular motion, minimum value of tension in the string
(A) zero (B) mg *(C) 15 mg (D) 10 mg

Passage-20

A particle is moved along the different paths OAC, OBC and ODC as shown in the figure. Path ODC is a parabola, $y = 4x^2$. The work done by a force $F = xy\hat{i} + x^2y\hat{j}$ on the particle along



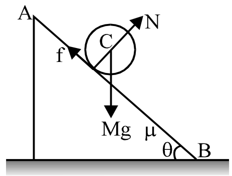
58. The path OAC is
*(A) 8 J (B) 2 J (C) 19/3 J (D) zero

59. The path OBC is
(A) 8 J *(B) 2 J (C) 19/3 J (D) zero

60. The path ODC is
(A) 8 J (B) 2 J *(C) 19/3 J (D) zero

Passage-21

Work energy theorem says "Work done by all the forces is equal to change in kinetic energy"
In the diagram, a disc of mass M and radius R is rolling without slipping on a rough inclined plane. Acceleration of disc along the plane is a. f is frictional force. Disc starts from rest from A. AB = ℓ. V = velocity of disc at B



61. Velocity of disc at B is
(A) $\sqrt{2g\ell\sin\theta}$ (B) $\sqrt{(2g\ell/3)\sin\theta}$
(C) $\sqrt{2g\ell}$ (*D) $\sqrt{(4g\ell/3)\sin\theta}$

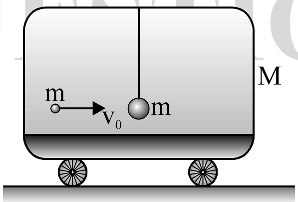
62. Kinetic energy of disc at B is
(A) $Mg\ell/2$ (*B) $(3M/4)v^2$
(C) $Mg\ell\sin\theta$ (D) $2Mg\ell\sin\theta/3$

63. Work done by frictional force (W_f) is
(A) zero (B) $-Mg\ell\sin\theta$
(C) $\frac{3}{4}MV^2 - Mg\ell\sin\theta$ (*D) both (A) and (C) are correct

IMPULSE AND MOMENTUM + COM

Passage - 22

A ball of mass m = 1 kg is hung vertically by a thread of length ℓ = 1.50 metre. Upper end of the thread is attached to the ceiling of a trolley of mass M = 4 kg. Initially, trolley is stationary and it is free to move along horizontal rails without friction. A shell of mass m = 1 kg, moving horizontally with velocity v_0 = 6 m/s, collides with the ball and gets stuck with it. As a result, thread starts to deflect towards right. (g = 10 m/s^2)



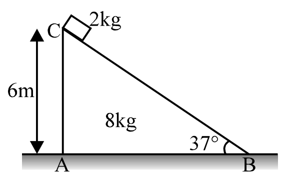
64. Velocity of combined mass 2m just after collision is
*(A) 3 m/sec (B) 6 m/sec (C) 1 m/sec (D) 1.5 m/sec

65. Velocity of the trolley, at the time of maximum deflection of the ball is
(A) 3 m/sec (B) 6 m/sec *(C) 1 m/sec (D) 1.5 m/sec

66. Maximum inclination of thread with the vertical is
(A) 30° *(B) 37° (C) 45° (D) 53°

Passage-23

A wedge ABC and a block are placed as shown in the figure. There is no friction at any surface.



67. Displacement of wedge ABC when 2 kg block reaches the point B

*(A) 1.6 m (B) 2 m (C) 1.5 m (D) 1.2 m

68. A person observes the 2 kg block from ground. He will observe that

(A) 2 kg block is moving at angle of 37° below horizontal

*(B) 2 kg block is moving at angle more than 37° below horizontal

(C) 2 kg blocks is moving at angle 90° with horizontal

(D) 2 kg block is moving along horizontal direction

69. Choose **incorrect** statement

(A) centre of mass of 2 kg block will move horizontally

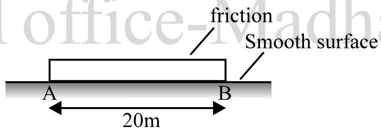
(B) centre of mass of 2 kg block will move vertically

*(C) centre of mass of whole system will move horizontally

(D) centre of mass of whole system will move vertically

Passage-24

A 100 kg uniform plank AB of length 20 m is placed on smooth horizontal surface. A boy of mass 25 kg is on edge A and start moving from A to B with constant velocity of 0.5 m/sec w.r.t the plank.



70. Displacement of boy w.r.t. ground when boy reaches to edge B

(A) 20 m *(B) 16 m (C) 15 m (D) 12 m

71. Velocity of plank

*(A) 10 cm/sec (B) 12.5 cm/sec (C) 25 cm/sec (D) 50 cm/sec

72. Choose **incorrect** statement

(A) Momentum of whole system is conserved

(B) momentum of boy can't remain conserved because of friction between plank and boy

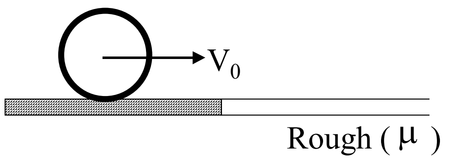
(C) momentum of plank can't remain conserved because of friction between plank and boy

*(D) position of centre of mass of whole system will change along positive x direction.

Passage-25

A ring of mass M and radius R sliding with a velocity v_0 suddenly enters into rough surface where the coefficient of friction is μ, as shown in fig

73. Choose the incorrect statement



(A) The friction does negative translational work

(B) The friction does positive rotational work

*(C) The net work done by friction is zero

(D) Friction force converts translational kinetic energy into rotational kinetic energy in rotational kinetic energy

74. Velocity of centre of mass of the ring when it starts rolling motion

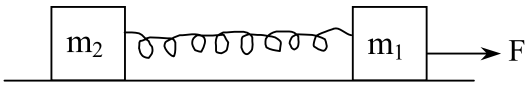
(A) $\frac{V_0}{4}$ *(B) $\frac{V_0}{2}$ (C) $\frac{V_0}{3}$ (D) $\frac{2V_0}{3}$

75. Linear distance moved by the centre of mass of the ring on the rough surface when it starts rolling is

*(A) $\frac{3}{8}\frac{V_0^2}{\mu g}$ (B) $\frac{2V_0^2}{8\mu g}$ (C) $\frac{V_0^2}{\mu g}$ (D) $\frac{5V_0^2}{8\mu g}$

Passage-26

Two blocks of masses m_1 and m_2 connected by an ideal spring of spring constant K are at rest on a smooth horizontal table. A constant horizontal force F acts on m_1. During the motion maximum elongation of the spring is x_0.



76. Both blocks m_1 and m_2 move with same velocity when the elongation of the spring is

(A) $\frac{m_2F}{2K(m_1+m_2)}$ (B) $\frac{m_2F}{K(m_1+m_2)}$ *(C) $\frac{2m_2F}{K(m_1+m_2)}$ (D) $\frac{4m_2F}{K(m_1+m_2)}$

77. Both m_1 and m_2 move with same acceleration when elongation of spring is

(A) $\frac{m_2F}{2K(m_1+m_2)}$ *(B) $\frac{m_2F}{K(m_1+m_2)}$ (C) $\frac{2m_2F}{K(m_1+m_2)}$ (D) $\frac{4m_2F}{K(m_1+m_2)}$

78. Select correct alternative

*(A) Velocity of center of mass is the same as that of m_1 and m_2 at the instant when elongation is x_0

(B) Velocity of center of mass is same as that of m_1 and m_2 at the instant when elongation of spring is $x_0/2$

(C) Velocity of center of mass can never be the as that of m_1 and m_2 at any instant

(D) Velocity of center of mass can become zero at an instant during the motion of system

Passage-27

Figure shows a dumbbell that consists of a mass-less rod and two particle size spheres. In figure (a) impulse is imparted perpendicular to the rod and in figure (b) impulse is imparted parallel to the rod. Answer following questions.

Figure (a) Figure (b)

79. Velocity of C.M. and angular velocity of system in figure (a) are respectively

(A) $\frac{\Delta\vec{p}}{M};\ \omega=\frac{\Delta p}{2Ml}$ *(B) $\frac{\Delta\vec{p}}{2M};\ \omega=\frac{\Delta p}{Ml}$ (C) $\frac{2\Delta\vec{p}}{M};\ \omega=\frac{\Delta p}{4Ml}$ (D) $\frac{\Delta\vec{p}}{2M};\ \omega=\frac{\Delta p}{2Ml}$

80. Velocity of C.M. and angular velocity of system in figure (b) are respectively

(A) $\frac{\Delta\vec{p}}{M};\ \omega=\frac{\Delta p}{2Ml}$ (B) $\frac{\Delta\vec{p}}{2M};\ \omega=\frac{\Delta p}{Ml}$ *(C) $\frac{\Delta\vec{p}}{2M};\ \omega=0$ (D) $\frac{\Delta\vec{p}}{2M};\ \omega=0$

81. Energies imparted to dumbbell in figure (a) and figure (b) are respectively

*(A) $\frac{(\Delta p)^2}{2M};\frac{(\Delta p)^2}{4M}$ (B) $\frac{(\Delta p)^2}{4M};\frac{(\Delta p)^2}{2M}$ (C) $\frac{(\Delta p)^2}{2M};\frac{(\Delta p)^2}{2M}$ (D) $\frac{(\Delta p)^2}{4M};\frac{(\Delta p)^2}{4M}$

GRAVITATION

Passage-28

Consider a geosynchronous communications satellite of mass m placed in an equitorial circular orbit of radius r_o. These satellites have an "apogee engine" which provides the thrusts needed to reach the final orbit.

Once an error by the ground controllers causes the apogee engine to be fired. The thrust happens to be directed towards the Earth and, despite the quick reaction of the ground crew to shut the engine off, an unwanted velocity variation Δv is imparted on the satellite. We characterize this boost by the parameter $\beta = \Delta v/v_o$. The duration of the engine burn is always negligible with respect to any other orbital times, so that it can be considered as instantaneous.

[**Hint:** Under the action of central forces obeying the inverse square law, bodies follow trajectories described by ellipses, parabolas or hyperbolas. In the approximation $m << M$ the gravitating mass M is at one of the focuses.

Where l is a positive constant named the semi-latus rectum and ε is the eccentricity of the curve. In terms of constants of motion:

$$l = \frac{L^2}{GMm^2} \quad \text{and} \quad \varepsilon = \left(1 + \frac{2EL^2}{G^2M^2m^3}\right)^{1/2}$$

where G is the Newton constant, L is the modulus of the angular momentum of the orbiting mass, with respect to the origin, and E is its mechanical energy, with zero potential energy at infinity . Suppose $\beta < 1$

82. Determine what kind of trajectory the satellite will take

(A) parabolic *(B) elliptic (C) hyperbolic (D) circular

83. Determine the distance r_{min} of the satellite to the earth centre

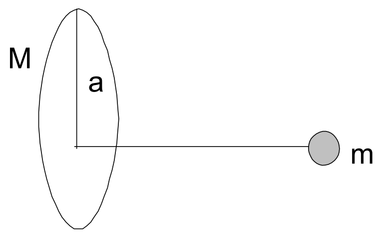
(A) $\frac{\beta r_o}{2}$ (B) $\frac{r_o}{1+\beta^2}$ *(C) $\frac{r_o}{1+\beta}$ (D) $\frac{r_0}{1+\sqrt{2\beta}}$

84. Determine the new time period of the satellite if $\left(\beta = \frac{1}{4}\right)$

(A) ≈ 20 hrs *(B) ≈ 26 hrs (C) ≈ 67.9 hrs (D) ≈ 48 hrs

Passage-29

A particle of mass m is placed at a distance x from the centre of ring along the line through the centre of the ring and perpendicular to its plane.



85. Gravitation potential energy of this system:

*(A) $-\frac{GMm}{\sqrt{a^2+x^2}}$ (B) $\frac{GMm}{a}$ (C) $\frac{GMm}{x}$ (D) None of these

86. The force , when x =0 :

(A) $\frac{GMm}{x^2}$ (B) $\frac{GMmx}{\sqrt{a^2+x^2}}$ (C) $\frac{GMmx}{\left(a^2+x^2\right)^{3/2}}$ *(D) Zero

87. If $x << a$, the particle of mass m will:
A) perform oscillatory motion
*B) perform S.H.M
C) Both
D) None

Passage-30

A satellite is revolving around the earth in a circular orbit of radius 'a' with velocity v_0. A particle is projected from the satellite in forward direction with relative velocity $v = \left(\sqrt{5/4} - 1\right) v_0$. In subsequent motion of the particle

88. Path traced by the satellite
(A) circle
*(B) ellipse
(C) straight line
(D) none of these

89. Its minimum distance from earth's centre.

*(A) 'a'
(B) $\frac{a}{2}$
(C) 3a
(D) $5\frac{a}{4}$

90. Its maximum distance from earth's centre.
(A) a
(B) $\frac{5a}{4}$
(C) infinity
*(D) $\frac{5a}{3}$

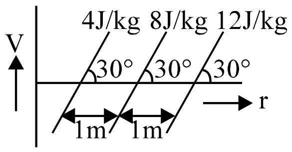
Passage-31

Gravitational force is a conservative and medium independent force. Its nature is attractive. Gravitational field intensity and gravitational potential gives information about gravitational field in vector and scalar form respectively. Actually gravitational field intensity is equal to the negative of the potential gradient. Potential energy is defined for only conservative force. It is also equal to the total energy in escaping condition. Gravitational potential is either negative or zero but can never be positive due to attractive nature of gravitational force.

91. A person brings a mass of 1 kg from infinity to a point A. Initially the mass was at rest but it moves with a speed of 2 m/s as it reaches A. The work done by the person on a mass is –3 J. The potential of A is :
(A) –3 J/kg
(B) –2 J/kg
*(C) –5 J/kg
(D) –7 J/kg

92. The gravitational potential inside a hollow sphere (mass M, radius R) at a distance r from the centre is :
(A) zero
*(B) $-\frac{GM}{R}$
(C) $-\frac{GM}{r}$
(D) $-\frac{GM}{r^2}$

93. Gravitational potential versus distance r graph is represented in figure. The magnitude of gravitational field intensity is equal to



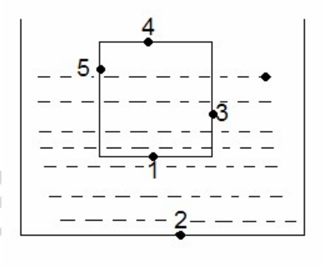
*(A) 8 Newton/kg
(B) 4 Newton/kg
(C) 6 Newton/kg
(D) 2 Newton/kg

ELASTICITY & FLUID MECHANICS

Passage-32

A liquid of density ρ is filled in a beaker of cross section A to a height H and a cylinder of ice of mass m base are 'a' is floating in it as shown in the figure. Take atmospheric pressure as P_0.

94. Pressure due to the weight of ice will be maximum at



*(A) 1 (B) 2 (C) both 1 & 2 (D) neither 1 or 2

95. Assume the temperature of liquid and surrounding is same, the rate of melting of ice is maximum at point

*(A)1 (B) 3 (C)4 (D) Both 4 & 5

96. If the whole system is falling freely under gravity then

a. $P_1 = P_2$ b. $P_2 = P_3$ c. $P_3 = P_4$ d. $P_1 = P_4$

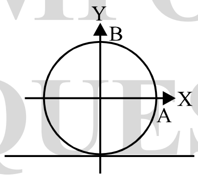
Choose the connect option

(A) only a and b are correct (B) only b and c are correct

(C) only a, b and c are correct *(D) all a,b ,c and d are correct.

Passage-33

A hollow sphere is completely filled with a liquid having a density ρ. The radius of sphere R. now sphere is pulled with a constant horizontal acceleration of g on a horizontal surface. Take centre of sphere as origin of co-ordinate system as shown in the figure.



97. Co-ordinate of point having minimum pressure is

(A) $\frac{R}{\sqrt{2}}, \frac{R}{\sqrt{2}}$ (B) $\frac{R}{2}, \frac{R}{2}$ *(C) $\frac{-R}{\sqrt{2}}, \frac{-R}{\sqrt{2}}$ (D) $\frac{-R}{2}, \frac{-R}{2}$

98 Co-ordinate of point having maximum pressure is

*(A) $\frac{R}{\sqrt{2}}, \frac{R}{\sqrt{2}}$ (B) $\frac{R}{2}, \frac{R}{2}$ (C) $\frac{-R}{\sqrt{2}}, \frac{-R}{\sqrt{2}}$ (D) $\frac{-R}{2}, \frac{-R}{2}$

99. Consider points A and B as shown in figure.

*(A) $P_A = P_B$ (B) $P_A > P_B$ (C) $P_B > P_A$ (D) $P_A = P_B = 0$

Passage -34

Poiseuille's equation for the flow rate relates the co-efficient of viscosity besides other parameters. According to Poiseuille's equation :

When a viscous liquid flows through a narrow tube, the volume rate of flow depends on pressure difference 'P' across the ends of tube as volume rate of flow

$$= \frac{\pi P r^4}{8 \eta \ell}$$

Here 'r' is radius of cross section of tube, 'ℓ' is length of the tube and η is coefficient of viscosity of liquid

Let us consider the situation regarding flow of blood in major arteries of the body. The coefficient of viscosity of blood at the body temperature is $4 \times 10^{-3}\, N - s/m^2$.

Answer the following:

100. Consider a major artery of radius 0.4cm. It carries blood at a flow rate 5 cc/sec. The pressure difference per meter in the artery is (density of mercury = 13.6gm/C.C)

(A) 9.6 mm of Hg (B) 3.2 mm of Hg *(C) 1.5 mm of Hg (D) 8.6 mm of Hg

101. If average value of arterial pressure is nearly 100mm of mercury, the answer obtained in previous question suggest that

(A) Viscous effects are significant in major arteries, so Bernoulli's equation is not applicable.

*(B) Viscous effects are insignificant in major arteries, so Bernoulli's equation is applicable

(C) It is not possible to judge the significance of viscous effects.

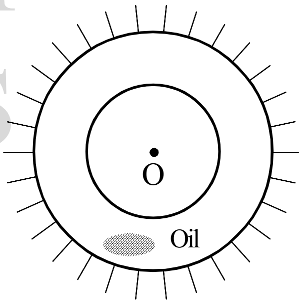
(D) Bernaulli's equation does not depend on viscosity.

102. Average blood pressure in the leg of a standing person 1m below the heart will be nearly (take average blood pressure at the level of heart as 100mm of Hg. Density of blood $\approx 10^3 Kg/m^3, g = 10 m/s^2$)

(A) 92 mm of Hg (B) 100 mm of Hg (C) 144 mm of Hg *(D) 174 mm of Hg

Passage-35

A cylinder 0.25 m in radius and 2 m length rotates coaxially inside a fixed cylinder of the same length and 0.3 m radius. Oil of viscosity 4.9 Ns/m^2 fills the space between the cylinders. A torque 4.9 N-m is applied to the inner cylinder. After constant velocity is attained



103. Calculate the velocity gradient at the walls of the inner cylinder

*(A) ≈ 127 /sec (B) ≈ 145 /sec

(C) ≈ 98 /sec (D) ≈ 110 /sec

104. What is the velocity gradient at the walls of the outer cylinder?

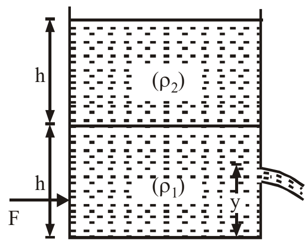
(A) ≈ 60 /sec (B) ≈ 70 /sec (C) ≈ 80 /sec *(D) ≈ 90 /sec

105. What is the power dissipated by fluid resistance ignoring end effects

(A) ≈ 50 W (B) ≈ 72 W *(C) ≈ 100 W (D) ≈ 125 W

Passage-36

A cylindrical tank having cross-sectional area A = 0.5 m^2 is filled with two liquids density $\rho_1 = 900\, kgm^{-3}$ and $\rho_2 = 600\, kg\, m^{-3}$, to a height h = 60 cm each as shown in figure. A small hole having area a = 5 cm^2 is made in right vertical wall at a height y = 20 cm from the bottom. Calculate



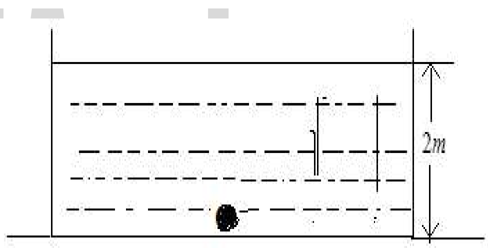
106. Velocity of efflux
*(A) $4\ ms^{-1}$ (B) $5\ ms^{-1}$ (C) $6\ ms^{-1}$ (D) $2\ ms^{-1}$

107. horizontal force F to keep the cylinder in static equlilibrium, if it is placed on a smooth horizontal plane, and
(A) 5.2 N *(B) 7.2 N (C) 6.2 N (D) 8.2 N

108. Maximum values of F to keep the cylinder in static equilibrium, if coefficient of friction between the cylinder and the plane is $\mu = 0.01$. ($g = 10\ m/s^2$)
(A) 62.2 N (B) 32.2 N *(C) 52.2 N (D) 42.2 N

Passage-37

A ball of mass 2 kg is released from the bottom of a tank filled with water (as an ideal liquid) up to 2m. On reaching the water surface the kinetic energy of the ball is found to be 16 J. The only forces involve here are gravity and upthrust. Upthrust is the force applied by liquid on the ball. (For any calculation g is taken as $10\ ms^{-2}$.)



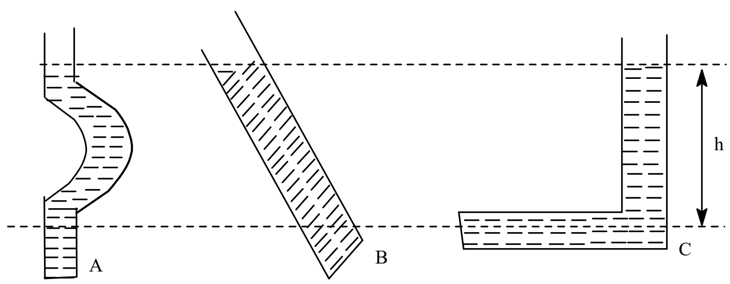
109. Choose the incorrect options
(A) work done by gravity forces is –40J.
(B) work done by up thrust is 56 J.
(C) work energy theorem can be applicable here
*(D) free falling observer (under gravity) observers that the work done by gravity on the ball is zero

110. What is the maximum height reached by the ball from the bottom of the tank ?
(A) more than 3.6 m if there is atmosphere above the water surface
(B) more than 3.6 m if there is no atmosphere above the water surface
*(C) less than 3.7 m if there is no atmosphere above the water surface
(D) the ball will not definitely come to the bottom again.

111. Select the correct options.
(A) If the system (tank+ball) is in the gravity free space the upthrust is non-zero but Gravitational force will be zero.
(B) From bottom to top the ball accelerates upwards with variable acceleration.
*(C) Under the circumstances mentioned in the passage the density of the material of ball is less than the density of water.
(D) Up thrust does not depend on gravitational acceleration.

Passage-38

The pressure at a point in a fluid in static equilibrium depends on the depth of that point but not on any horizontal dimension of the fluid or its container. When a body floats in a fluid, the magnitude F_g of the gravitational force on the body is equal to weight mg of the fluid that has been displaced by the body.



112. Fig shows four containers of same oil. Rank them according to pressure at depth h, greatest first.

(A) A, C, B (B) B, A, C (C) C, A, B *(D) All tie

113. A penguin floats first in a liquid of density ρ_0 then in a fluid of density $0.95\rho_0$ and then in a fluid of density $1.1\rho_0$ .Rank the densities according to the magnitude of buoyant force on the penguin greatest first.

(A) $1.1\rho_0$, ρ_0, $0.95\rho_0$ (B) $0.95\rho_0$, ρ_0, $1.1\rho_0$

(C) ρ_0, $0.95\rho_0$, $1.1\rho_0$ *(D) All tie

114. In the above problem rank the densities according to the amount of fluid displaced by the penguin, greatest first.

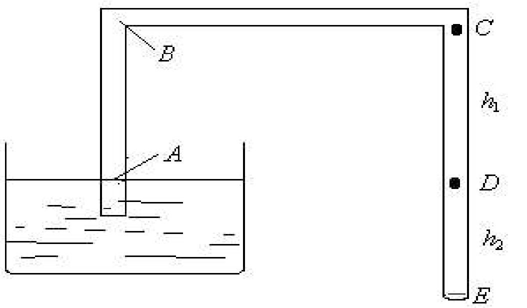
(A) $1.1\rho_0$, ρ_0, $0.95\rho_0$ *(B) $0.95\rho_0$, ρ_0, $1.1\rho_0$

(C) ρ_0, $0.95\rho_0$, $1.1\rho_0$ (D) All time

Passage-39

A siphon in action is shown in the diagram. A is the point inside the siphon at the water level of container. D is at the same level as A. The cross section of the siphon is uniform. E end is open to atmosphere.

(Given $AB = CD = h_1$; $DE = h_2$).



Answer the following questions

115. As water flow is established inside the siphon

(A) $P_B > P_C$ *(B) $P_B = P_C$ (C) $P_B < P_C$

(D) pressure at B and C can not be predicted.

116. As water flows through the siphon the flow velocity is dependent on

(A) h_1 only *(B) h_2 only (C) h_1 and h_2 both (D) neither h_1 nor h_2

117. Pressure at A is

(A) Greater than atmospheric (B) equal to atomospheric

*(C) less than atomospheric (D) can't be predicted

WAVES

Passasge-40

A stretched wire, 0.90 m long, resonates with a tone whose wavelength is 0.72 m.

118. The number of the harmonic, for this resonant wavelength is :

(A) 2 (B) 4 (C) 6 *(D) 5

119. The distance between a node and an adjacent antinode, in the standing wave pattern in the pipe, in SI units, is closest to

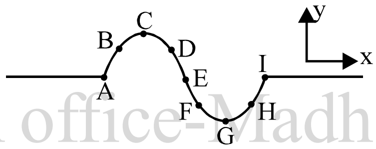
*(A) 0.18 (B) 0.36 (C) 0.27 (D) 0.22

120. The wavelength of the next higher overtone in this pipe, in SI units, is closest to

(A) 0.40 (B) 0.58 (C) 0.36 *(D) 0.51

Passasge-41

A progressive wave pulse is generated on a string is traveling in –ve X direction as shown in figure.



121. Kinetic energy is maximum for the particle

(A) D *(B) E (C) F (D) G

122. Potential energy is maximum for the particle

(A) D *(B) E (C) F (D) G

123. Particle B and D are moving respectively

(A) ↑↑ (B) ↓↓ *(C) ↑↓ (D) ↓↑

Passasge-42

A standing wave exists in a string of length 150 cm. and is fixed at both ends. The displacement amplitude of a point at a distance of 10cm from one of the ends is $5\sqrt{3}$ mm. The distance between the two nearest points, with in the same loop and having displacement amplitude equal to $5\sqrt{3}$ mm, is 10

124. The maximum displacement amplitude of the particle in the string _____

*(A) 10mm (B) $20/\sqrt{3}$ mm (C) $10\sqrt{3}$ mm (D) 20 mm

125. The mode of vibration of the string i.e. the overtone produced –

(A) 2 (B) 3 *(C) 4 (D) 6

126. At what minimum distance from one end, is the potential energy of string zero when this end has maximum P.E. –

(A) $10\sqrt{3}$ cm *(B) 15 cm (C) 20 cm (D) 30 cm

Passasge-43

One observe that the Doppler effect is not associated with wave motion only but it is a more general phenomenon and the phenomenon described below is known as "classical Doppler effect" "An Indian fighter plane flying at a velocity of 300 m/s on the fighter with a gun which shoots at a rate of 40 rounds/sec with a muzzle velocity of 1200 m/sec. The shots are aimed at a Pakistani's plane flying at a velocity of 200 m/sec. Due to relative motion between the two planes the rate at which bullet hits the pakistani plane is different from the rate at which it is shoot from the Indian plane. Find the rate (in round per sec) at which the bullets hits the Pakistani plane

127. When the two planes move in the same direction and the target plane is in front of the Indian plane
(A) 36.67 *(B) 43.33 (C) 56.67 (D) 23.33

128. When the target plane is behind the shooting plane but moving in the same direction
*(A) 36.67 (B) 43.33 (C) 56.67 (D) 23.33

129. When the two planes move towards one another
(A) 36.67 (B) 43.33 *(C) 56.67 (D) 23.33

Passage-44

A standing wave exists in a string of length 150 cm. and is fixed at both ends. The displacement amplitude of a point at a distance of 10cm from one of the ends is $5\sqrt{3}$ mm. The distance between the two nearest points, with in the same loop and having displacement amplitude equal to $5\sqrt{3}$ mm, is 10

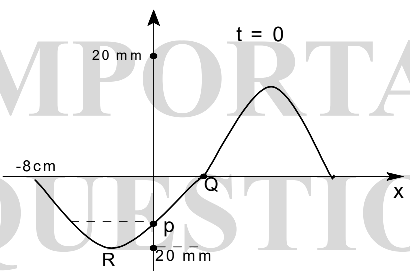
130. The maximum displacement amplitude of the particle in the string
(*A) 10mm (B) $20/\sqrt{3}$ mm (C) $10\sqrt{3}$ mm (D) 20 mm

131. The mode of vibration of the string i.e. the overtone produced
(A) 2 (B) 3 (*C) 4 (D) 6

132. At what minimum distance from one end, is the potential energy of string zero when this end has maximum P.E. –
(A) $10\sqrt{3}$ cm (*B) 15 cm (C) 20 cm (D) 30 cm

Passage-45

A plane sinusoidal sound wave is propagating along +x-axis with speed 120cm/s and having frequency 5Hz. At $t = 0$, the displacement S of the particles is plotted as function of position x as shown in figure.



133. In figure, what is the distance of point Q from the origin
(A) 2 cm (B) 3 cm *(C) 4 cm (D) 1cm

134. Velocity of the particle P in figure is
(A) $10\sqrt{3}\pi$ cm/s upward (B) 10π cm/s downward
(C) $10\sqrt{3}\pi$ cm/s right *(D) 10π cm/s left

135. What is the velocity of particle R in figure when particle P reach its mean position?
(A) $10\sqrt{3}\pi$ cm/s, upward (B) 10π cm/s, downward
*(C) $10\sqrt{3}\pi$ cm/s, right (D) 10π cm/s, left

Passage-46

A plane longitudinal wave having frequency 500 rad/s is traveling in positive direction in medium of density 160g/m^3 and of bulk modulus 4 x 10^4 N/m^2. The loudness at a point in the medium is observed to be 20dB.

136. The speed of wave is
*(A) 500 m/s (B) 250 m/s (C) 100 m/s (D) 50 m/s

137. Displacement amplitude of wave (in nm) is
(A) 2.0 *(B) 3.0 (C) 5.0 (D) 8.0

138. If wave refract into an another medium with refraction angle is twice of the incidence angle which is very very small, then the equation of wave in second medium in SI units is
*(A) $P = \left(1.2 \text{ x } 10^{-4}\right) Sin\left[500\left(t-2x\right)\right]$ (B) $P = \left(3.0 \text{ x } \mathbf{10^{-9}}\right) Sin\left[500\left(t-x\right)\right]$
(C) $p = \left(3.0 \text{ x } 10^{-9}\right) Sin\left[500\left(t-2x\right)\right]$ (D) $P = \left(1.2 \text{ x } 10^{-4}\right) Sin\left[500\left(t-x\right)\right]$

Passage-47
A string with linear mass density 4×10^{-3} kg/m is under tension of 360 N and is fixed at both ends. One of its resonance frequencies are 375 Hz The next higher frequency is 450 Hz.

139. What is the length of the string ?
*(A) 2 m (B) 4 m (C) 0.5 m (D) 1 m

140. When the string resonance frequency is close to 1000 Hz, in which overtone is it vibrating?
(A) 14^{th} (B) 13^{th} (C) 11^{th} *(D) 12^{th}

141. Which of the following equation can not represent the equation of wave in the string?
(A) 0.01 sin2πx cos600πt (B) 0.25 sin(9πx/2) sin(1080π)t
*(C) 0.05 sin(πx/4) cos75πt (D) 0.01 sinπx sin300πt

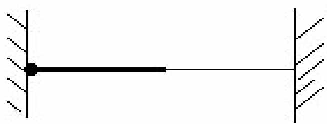
Passage-48
A person standing between a pair of tall and wide cliffs claps his hands. He hears the first two echoes at 2 second and 3 second respectively. It the speed of sound is 330 ms^{-1}, then

142. The separation between the cliffs is
*(A) 825 m (B) 800 m (C) 725 m (D) 600 m

143. The time of third echo is
(A) 3 sec *(B) 5 sec (C) 2.5 sec (D) 2 sec

144. Choose the correct statement
(A) after the second echo, none is formed *(B) every third echo is the loudest
(C) every second echo is the loudest (D) none of the above

Passage-49
A String of 3m length and linear mass density $25 \times 10^{-3} kg/m$ is tied at one end to a fixed rigid support and to a light thread at the other end as shown. The tension in the string is 160 N. The string oscillates in its first overtone and sets a closed organ pipe of length 3 m into vibration in its fundamental mode.
(in which some unknown gas is filled)
Answer the following questions:



145. The velocity of sound wave in the string is
(A) 40 m/s (B) 100 m/s (C) 120 m/s *(D) none of these

146. The frequency with which the wire oscillates is
(A) 30 Hz (B) 40 Hz *(C) 20 Hz (D) none of these

147. The velocity of sound in the unknown gas is
(A) 340 m/s *(B) 240 m/s (C) 200 m/s (D) none of these

Passage-50

The displacement of the medium in a sound wave is given by the equation

$$y_1 = A\cos(ax + bt)$$

where A, a and b are positive constants. The wave is reflected by an obstacle situated at x = 0. The intensity of the reflected wave is 0.64 times that of the incident wave.

148. The equation for the reflected wave is
(A) $y_1 = 0.8A\sin(bt - ax)$ (B) $y_1 = 0.8A\cos(bt - ax)$
*(C) $y_1 = -0.8A\cos(bt - ax)$ (D) $y_1 = -0.8A\sin(bt - ax)$

149. In the resultant wave formed after reflection, the maximum and minimum values of the particle speed in the medium.
*(A) v_{max} = 1.8 Ab and v_{min} = 0.2 Ab (B) v_{max} = 1.6 Ab and v_{min} = 0.1 Ab
(C) v_{max} = 1.4 Ab and v_{min} = 0.1 Ab (D) v_{max} = 1.2 Ab and v_{min} = 0.2 Ab

150. Express the resultant wave as a superposition of a standing wave and a travelling wave
*(A) $y = 2A\sin bt\sin ax + 0.2A\cos(bt - ax)$ (B) $y = 2A\cos bt\cos ax + 0.2A\cos(bt - ax)$
(C) $y = 2A\cos bt\cos ax + 0.2A\sin(bt - ax)$ (D) $y = 2A\sin bt\sin ax + 0.2A\sin(bt - ax)$

Passage-51

A progressive and a stationary simple harmonic wave each has the same frequency of 235 Hz and the same velocity of 30 m/s. Calculate

151. The phase difference between two vibrating points on the progressive wave which are 10 cm apart.
*(A) $\frac{5}{3}\pi$ (B) $\frac{5}{6}\pi$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{12}$

152. Distance between nodes in stationary wave
(A) $\frac{25}{3}$ m (B) 12 cm *(C) 6 cm (D) $\frac{5}{3}$ m

153. Equation of stationary wave of its amplitude is given by
(A) $0.01\sin\left(\frac{5\pi x}{3}\right)\sin(500\pi t)$ *(B) $0.01\cos\left(\frac{5\pi x}{3}\right)\sin(500\pi t)$
(C) $0.02\cos\left(\frac{5\pi x}{3}\right)\cos(500\pi t)$ (D) $0.02\cos\left(\frac{5\pi x}{3}\right)\sin(500\pi t)$

Passage-52

If the tension in a string is changed by 21 %, fundamental frequency of the string changes by 15 Hz.

154. Original fundamental frequency is about
*(A) 150 Hz. (B) 100 Hz (C) 120 Hz (D) 50 Hz

155. Velocity of propagation changes by nearly
(A) 5% *(B) 10% (C) 20% (D) 25 %

156. Fundamental wavelength changes by nearly
(A) 5% (B) 10% (C) 20% *(D) 0%

SIMPLE HARMONIC MOTION

Passage-53

A particle moves along the x-axis according to the law $S = a\sin^2\left(wt - \pi/4\right)$

157. The amplitude of the oscillation is

(A) a *(B) $\frac{a}{2}$ (C) $\frac{3a}{2}$ (D) $2a$

158. The time period of oscillations is

(A) $\frac{2\pi}{w}$ *(B) T/W (C) $\frac{3\pi}{2W}$ (D) $\frac{\pi}{2W}$

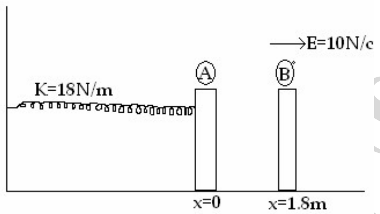
159. The velocity of the particle (u), as a function of the co-ordinate $'x'$ is

*(A) $u^2 = 4w^2x(a-x)$ (B) $u^2 = 2w^2x(a-x)$

(C) $u^2 = w^2x(a-x)$ (D) $u^2 = 8w^2x(a-x)$

Passage-54

In the figure shown $m_A = m_B = 1kg$ block-A is neutral while block-B carries charge -1c. Sizes of A and B are negligible. Block-B is released from rest at a distance 1.8m from A. Initially spring is neither compressed nor stretched.



160. If collision between A and B is perfectly inelastic, then velocity of the combined mass just after collision is

(A) $6m/s$ *(B) $3m/s$ (C) $9m/s$ (D) $12m/s$

161. Equilibrium position of the combined mass is at x=

(A) $\frac{-2}{9}m$ (B) $\frac{-1}{3}m$ *(C) $\frac{-5}{9}m$ (D) $\frac{-7}{9}m$

162. The amplitude of oscillation of the combined mass

(A) $\frac{2}{3}m$ (B) $\frac{\sqrt{124}}{3}m$ (C) $\frac{\sqrt{72}}{9}m$ *(D) $\frac{\sqrt{106}}{9}m$

Passage-55

Two blocks of masses m and 2m rest on a frictionless horizontal surface. They are connected by an ideal spring of relaxed length l and stiffness constant k. By means of a massless thread connecting the blocks the spring is held compressed to a length $l/2$. The whole system is moving with speed v in a direction perpendicular to the length of the spring. The thread is then burnt. Answer the following questions in terms of l, k, m, v

163. Before burning, the ratio of kinetic energy to potential energy is :

*(A) $\frac{12mv^2}{kl^2}$ (B) $\frac{mv^2}{12kl^2}$ (C) $\frac{12kl^2}{mv^2}$ (D) $\frac{kl^2}{12mv^2}$

164. After burning the string the speed of centre of mass is :

(A) less than v (B) greater than v *(C)equal to v (D) equal to $\frac{2}{\sqrt{3}}v$

165. The time period of oscillation of the system is :

*(A) $2\pi\sqrt{\frac{2m}{3k}}$ (B) $2\pi\sqrt{\frac{3m}{2k}}$ (C) $2\pi\sqrt{\frac{3m}{k}}$ (D) $\pi\sqrt{\frac{2m}{3k}}$

Passage-56

A car is driven at constant speed on a rough road. The shock absorbers vibrate vertically. As we know, if the displacement varies sinusoidally, the oscillation will be a simple harmonic. When there is matching between the disturbance and the oscillating shockers, we call it as resonance.

Consider the mass of the car and its contents M = 1500 kg with a man of mass m = 75 kg, rising vertically S = 0.1 m due to the peaks separated by $\lambda = 10$ m equally and the speed of the car being v and time period being T. Then:

166. The frequency of oscillation of the car is proportional to :

*(A) $\sqrt{\frac{mg}{MS}}$ (B) $\sqrt{\frac{mg}{d}}$ (C) $\sqrt{\frac{3mg}{2MS}}$ (D) $\sqrt{\frac{mg}{Md}}$

167. Critical speed at maximum displacement is (in ms^{-1}) :

(A) 2.56 *(B) 3.56 (C) 5.26 (D) 3.76

168. The nature of oscillation of shocker will be :

*(A) damped oscillations
(B) simple harmonic oscillations of constant amplitude
(C) periodic with increasing amplitude
(D) information given in the passage are insufficient

Passage-57

The equation of a particle executing SHM is $\frac{d^2x}{dt^2} = -\omega^2 x$. where $\omega = \frac{2\pi}{\text{time period}}$. The velocity of particle is maximum when it passes through mean position and its accleration is maximum at extremeposition. The displacement of particle is given by $x = A\sin(\omega t + \theta)$ Where θ-initial phase of motion. A-Amplitude of motion and T-Time period

169. The time period of pendulum is given by the equation $T = 2\pi\sqrt{\frac{l}{g}}$. Here $\frac{d^2x}{dt^2}$ is :

*(A) $\frac{-l}{g}x$ (B) $\frac{g}{l}x$ (C) $\frac{1}{g}x$ (D) $\frac{-g}{l}x$

170. The average velocity during motion of particle from one extreme point to the other extreme point :

*(A) $\frac{2v}{\pi}$ (B) $\frac{2A}{t}$ (C) $\frac{3A}{t}$ (D) $\frac{v}{\pi}$

171. The accleration is half of its maximum value at an amplitude of

(A) $\frac{A}{\sqrt{2}}$ (B) $\frac{\sqrt{3}A}{2}$ (C) $\frac{A}{\sqrt{3}}$ *(D) $\frac{A}{2}$

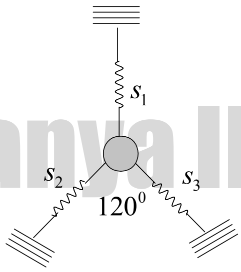
Passage-58

A spring of force constant K is attached with a mass executes with S.H.M has a time period T. It is cut into three equal parts

172. If the springs are connected in parallel then the time period of the combination for the same mass is

(A) T *(B) $\frac{T}{3}$ (C) $\frac{T}{9}$ (D) $\frac{2T}{3}$

173. If the springs are attached to the load as shown in figure then effective spring constant is



(A) $\frac{K}{2}$ (B) $\frac{K}{3}$ (C) $\frac{5K}{3}$ *(D) $\frac{3K}{2}$

174. The frequency of arrangement when the springs are connected in series to the parallel combinaiton for the same load is

(A) same in both cases (B) More in series *(C) More in parallel (D) can not say

Passage-59

Two light vertical springs 1 and 2 of stiffness K and 2K are connected by a l ight horizontal rod. A bead of chewing gum of mass 'm' is placed gently at the mid point of the rod and lowered slowly.

175. The energy of oscillation of the bead for an amplitude y_0 is equal to (x_0 = maximum displacement of the bead in the process of lowering)

(A) mgx_0 (B) $mg(x_0 + y_0)$ (C) $\frac{3}{2}Ky_0^{\,2}$ *(D) $\frac{4}{3}Ky_0^{\,2}$

176. The period of SHM of the bead is

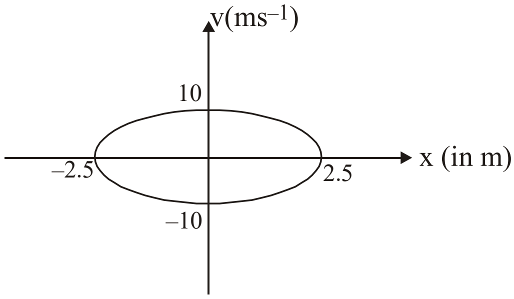
*(A) $2\pi\sqrt{\frac{3m}{8K}}$ (B) $2\pi\sqrt{\frac{m}{3K}}$ (C) $3\pi\sqrt{\frac{2m}{K}}$ (D) None of these

177. The ratio of magnitude of work done by the external agent to that by gravity in lowering the bead till it remains at rest is

(A) 1 : 1 *(B) 1 : 2 (C) 2 : 1 (D) 1 : 4

Passage-60

The figure shows graph between velocity 'v' and displacement 'x' from mean position of a particle performing simple harmonic motion



178. The time period of motion of particle is :

(A) $\frac{\pi}{4}$ s *(B) $\frac{\pi}{2}$ s (C) π s (D) 2π s

179. The maximum acceleration of particle is :

(A) $10ms^{-2}$ (B) $20\ ms^{-2}$ (C) $30\ ms^{-2}$ *(D) $40\ ms^{-2}$

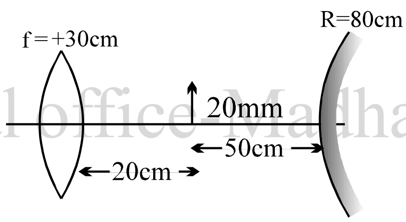
180. The velocity of particle when it is at a distance 1.5m from mean position is :

(A) $4ms^{-1}$ *(B) $8\ ms^{-1}$ (C) $12\ ms^{-1}$ (D) $16\ ms^{-1}$

OPTICS

Passage-61

An optical system comprises in turn, from left to right : an observer, a lens of focal length + 30 cm, an object 20 mm high, and a convex mirror of radius 80 cm. The object is between the lens and the mirror, 20 cm from the lens and 50 cm from the mirror. The observer views the image formed first by reflection and then by refraction



181. In figure, the character of the intermediate image formed by the mirror, with respect to the erect object, is

*(A) virtual and diminished (B) real and erect
(C) indeterminate (D) virtual and inverted

182. In figure, the position of the final image, measured from the mirror, in cm, is closest to

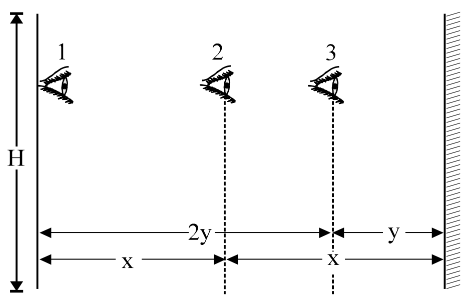
(A) 90 (B) 138 (C*) 114 (D) 126

183. In figure, the character of the final image, with respect to the erect object, is

(A) real and erect (B) indeterminate
*(C) real and inverted (D) virtual and erect

Passage-62

A man is standing in a room and a plane mirror is fixed on the wall in front of him. The height of the wall is H. Man has to see the complete image of wall behind him at a time.



184. When the man is standing near wall i.e. at position 1, the minimum length of the mirror required is

(A) H *(B) H/2 (C) H/3 (D) H/4

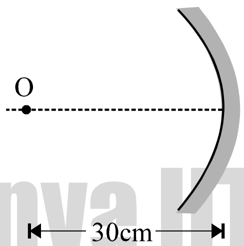
185. When the man is standing at the middle of the room i.e. at position 2, the minimum length of the mirror required is

(A) H (B) H/2 *(C) H/3 (D) H/4

186. When the man is at position 3, the minimum length of the mirror required is
(A) H (B) H/2 (C) H/3 *(D) H/4

Passage-63

An object is present on the principal axis of a concave mirror at a distance 30 cm from it. Focal length of mirror is 20 cm.



187. Image formed by mirror is
*(A) At a distance 60 cm in front of mirror
(B) At a distance 60 cm behind the mirror
(C) At a distance 12 cm in front of mirror
(D) At a distance 12 cm behind the mirror.

188. If object starts moving with 2 cms^{-1} along principal axis towards the mirror then,
*(A) image starts moving with 8 cms^{-1} away from the mirror
(B) image starts moving with 8 cms^{-1} towards the mirror
(C) image starts moving with 4 cms^{-1} towards the mirror
(D) image starts moving with 4 cms^{-1} away from the mirror

189. If object starts moving with 2 cms^{-1} perpendicular to principal axis above the principal axis then,
*(A) image moves with velocity 4 cms^{-1} below the principal axis
(B) image moves with velocity 4 cms^{-1} above the principal axis
(C) image moves with velocity 8 cms^{-1} below the principal axis
(D) image moves with velocity 8 cms^{-1} above the principal axis

Passage-64

Real images of an object are formed on the screen for two positions of a lens separated by a distance 60 cm.

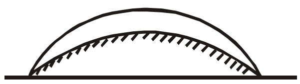
190. If the distance between object and screen is 180 cm then, the focal length of the lens is :
*(A) 40 cm (B) 25 cm (C) 14 cm (D) 20 cm

191. The ratio between the sizes of the two images will be :
(A) 1/3 (B) 3 *(C) 1/4 (D) 1/2

192. The images in the two cases have sizes 0.4 cm and 0.9 cm. The size of the object will be :
(A) 0.36 cm *(B) 0.6 cm (C) 1.5 cm (D) 0.9 cm

Passage-65

The concave surface of a thin concavo-convex lens of index 1.5 has radius of curvature 50 and 10 cm respectively. The concave side is silvered and placed on a horizontal surface as shown



193. Focal length of lens (without silvering) is :
*(A) 25 cm (B) 20 cm (C) 10 cm (D) 5 cm

194. Focal length of the silvered lens is :
(A) 30 cm *(B) 25 cm (C) 15 cm (D) 10 cm

195. The object distance at which its image coincides with itself is :
*(A) 50 cm (B) 60 cm (C) 30 cm (D) 20 cm

Passage-66

An object 25 cm high is placed in front of a convex lens of focal length 30cm. If the height of image formed is 50cm.

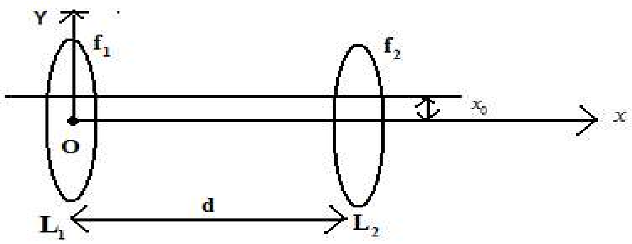
196. The image distance is, if the image is real and inverted;
*(A) 90 cm (B) 45 cm (C) 30 cm (D) 25cm

197. If the image is erect and virtual then the image distance is
(A) 90 cm (B) 45 cm *(C) -30 cm (D) –60 cm

198. If the image is erect and virtual then distance between object and image is
(A) 25 cm *(B) 15 cm (C) 40 cm (D) 60 cm

Passage-67

Two thin convex lenses of focal lengths f_1 and f_2 are separated by a horizontal distance 'd' where $(d < f_1 \text{ and } d < f_2)$ and their optical centers are displaced vertically by x_0. Taking the origin at the centre of lens L_1, consider the following cases when a parallel beam of paraxial rays are coming from left.



199. For the lens L_2, object distance is
(A) $f_2 - f_1$ (B) $f_2 + d$ (C) $f_1 + d$ *(D) $(f_1 - d)$

200. x coordinates of the final image is

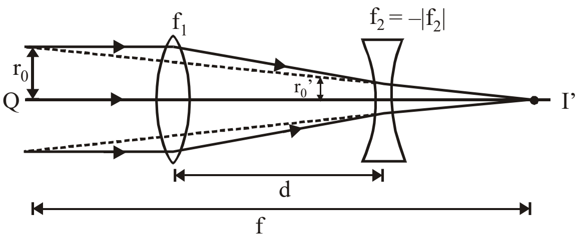
*(A) $\dfrac{d(f_1 - d) + f_1 f_2}{(f_1 + f_2 - d)}$ (B) $\dfrac{d(f_1 + d)}{f_1 + f_2 - d}$ (C) $f_1 + f_2$ (D) $\dfrac{(f_1 + f_2)d}{f_1 - f_2 + d}$

201. Y coordinate of the final image is given by:

*(A) $x_0\left[\dfrac{f_1 - d}{(f_1 + f_2 - d)}\right]$ (B) x_0 (C) $\dfrac{x_0(f_1 + d)}{f_1 + f_2}$ (D) $\dfrac{x_0(f_1 + f_2)}{(f_1 + f_2 - d)}$

Passage-68

The following figure shows a simple version of a zoom. The converging lens has a focal length f_1 and the diverging lens has focal length $f_2 = -|f_2|$. The two lens are separated by a variable distance d that is always less than f_1, also the magnitude of the focal length of the diverging lens satisfies the inequality $|f_2| > (f_1 - d)$.



If the rays that emerge from the diverging lens and reach the final image point are extended backward to the left of the diverging lens, they will eventually expand to the original radius r_0 at the same point Q. To determine the effective focal length of the combination lens consider a bundle of parallel rays of radius r_0 entering the emerging lens.

202. At the point where ray enters the diverging lens, the radius of the ray bundle decreases to

*(A) $r_0'=\left(\frac{f_1-d}{f_1}\right)r_0$ (B) $r_0=\left(\frac{f_1-d}{f_1}\right)r_0'$ (C) $r_0'=\left(\frac{f_1-f_2}{f_1}\right)r_0$ (D) $r_0'=r_0$

203. To the right of the diverging lens the final image I' is formed at a distance S_2' given by

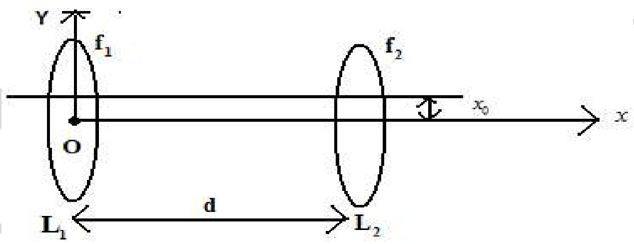
(A) $\frac{(f_1-f_2)d}{f_1-f_2+d}$ *(B) $\frac{(f_1-d)\ f_2}{|f_2|+d-f_1}$ (C) $\frac{f_1-f_2+d}{f_1-f_2}$ (D) $\frac{(d-f_1)f_2}{f_1-f_2-d}$

204. The effective focal length f is given by

*(A) $\frac{f_1|f_2|}{|f_2|-f_1+d}$ (B) $\frac{f_1f_2}{f_1-f_2-d}$ (C) $\frac{f_1}{f_2-f_1+d}$ (D) $\frac{f_2}{f_1+f_2+d}$

Passage-69

Two thin convex lenses of focal lengths f_1 and f_2 are separated by a horizontal distance 'd' where $(d < f_1 \text{ and } d < f_2)$ and their common centers are displaced vertically by x_0. Taking the origin at the centre of lens L_1, consider the following cases when a parallel beam of paraxial rays are coming from left.



205. For the lens L_2, object distance is

(A) f_2-f_1 (B) f_2+d (C) f_1+d *(D) (f_1-d)

206. x coordinates of the final image is

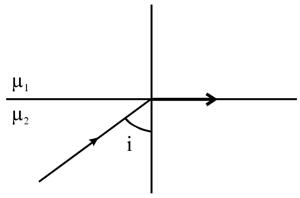
*(A) $\frac{d(f_1-d)+f_1f_2}{(f_1+f_2-d)}$ (B) $\frac{d(f_1+d)}{f_1+f_2-d}$ (C) f_1+f_2 (D) $\frac{(f_1+f_2)d}{f_1-f_2+d}$

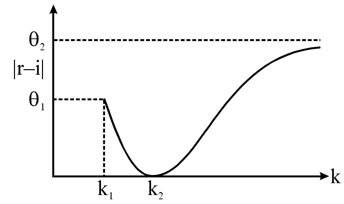
207. Y coordinate of the final image is given by:

*(A) $x_0\left[\frac{f_1-d}{(f_1+f_2-d)}\right]$ (B) x_0 (C) $\frac{x_0(f_1+d)}{f_1+f_2}$ (D) $\frac{x_0(f_1+f_2)}{(f_1+f_2-d)}$

Passage-70

The figure shows a ray incident at an angle $i = \pi/3$. If the plot drawn shows the variation of $|r - i|$ versus $\frac{\mu_1}{\mu_2} = k$, (r = angle of refraction)





208. The value of k_1 is

(A) $\frac{2}{\sqrt{3}}$ (B) 1 (C) $\frac{1}{\sqrt{3}}$ *(D) $\frac{\sqrt{3}}{2}$

209. The value of θ_1 is

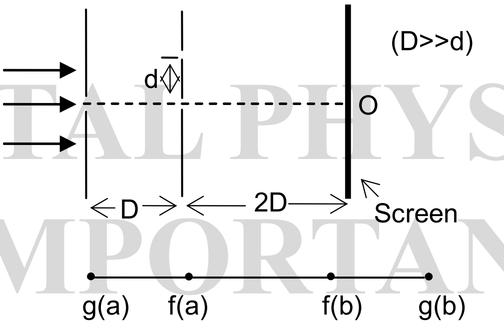
(A) $\pi/3$ (B) $\pi/2$ *(C) $\pi/6$ (D) zero

210. The value of k_2 is

*(A) 1 (B) 2 (C) 1/2 (D) none

Passage-71

In the arrangement shown in the fig, the distance D is large compared to the separation d between the slits. Find :



211. The minimum value of d for which there is a dark fringe at the point O.

(A) $\sqrt{\lambda D/2}$ *(B) $\sqrt{D\lambda/3}$

(C) $\sqrt{\lambda D}$ (D) cannot be calculated

212. The position of first bright fringe for the minimum value of d is

(A) d/2 below (B) d above

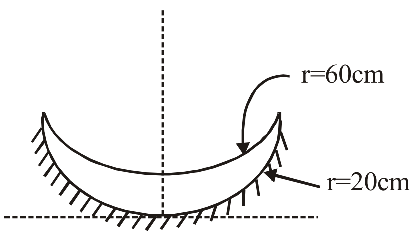
*(C) 3d/2 below (D) 3d/2 above

213. The fringe width is

(A) $\frac{3D\lambda}{4d}$ (B) $\frac{3D\lambda}{2d}$ (C) $\frac{D\lambda}{d}$ *(D) $\frac{2D\lambda}{d}$

Passage-72

The convex surface of a thin concavo–convex lens of glass of refractive index 1.5 has a radius of curvature of 20 cm. The concave surface has a radius of curvature of 60 cm. The convex side is silvered and placed on a horizontal surface as shown in the figure.



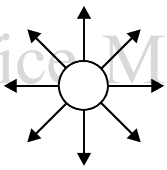
214. The focal length of the combination has the magnitude
(A) 8.6 cm *(B) 7.5 cm (C) 1.5 cm (D) 15 cm

215. The combination behaves like
*(A) a concave mirror (B) a convex mirror
(C) a concave lens (D) a convex lens

216. A small object is placed on the principal axis of the combination, at a distance of 30 cm in front of the mirror. The magnification of the image is
(A) $-\frac{1}{3}$ (B) $\frac{3}{4}$ (C) 5 *(D) none of these

Passage-73

On the basis of the phenomenon of polarisation, it was established that light is a form of transverse wave motion. Since light is an electromagnetic wave, it consists of electric vector $\vec{E}$ and magnetic vector $\vec{H}$ varying periodically in phase, at right angles to each other and also perpendicular to the direction of propagation.



In ordinary light, the electric vector vibrates in all planes with equal probability at right angles to the direction of propagation. Hence the ordinary light also called unpolarised light can be represented by a star as shown.
Hence unpolarised light is symmetrical about the direction of propagation. By the phenomenon of reflection, light can be polarised. The polarised light has vibrations confined only to a single line in a plane perpendicular to the direction of propagation and is called plane polarised.
To polarise light by reflection we make use of Brewster's law which states that the tangent of the angle of polarisation (p) is numerically equal to the refractive index μ of the medium. The device that polarises unpolarised light is called the polariser and the device that analyses the polarised light is called the analyser.
The amplitude of a plane polarised light incident on the analyser can be resolved into two components, one parallel to the plane of transmission of the analyser and the other perpendicular to it. The parallel component, namely, (a cosθ) is transmitted through the analyser.

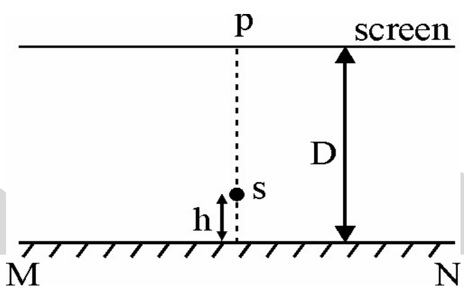
217. At the angle of polarisation, the angle of inclination between the reflected and refracted rays is :
(A) $\frac{\pi}{8}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ *(D) $\frac{\pi}{2}$

218. If in an unpolarised light $\vec{E} = 2\hat{i} + 3\hat{j}$ and $\vec{H} = 3\hat{i} - 2\hat{j}$, then the direction of propagation is given by :
*(A) $-13\vec{k}$ (B) $-10\hat{j}$ (C) $-6\hat{i} + 6\hat{k}$ (D) $5\hat{i} - 4\hat{k}$

219. An analyser examines two adjacent plane-polarised beams A and B whose planes of vibration are mutually perpendicular. In one position of the analyser, beam B shows zero intensity. From this position, a rotation of 30° shows the two beams as matched in intensity. The intensity ratio $\frac{I_A}{I_B}$ of the two beams is :
(A) $\sqrt{3}$ *(B) $\frac{1}{3}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{5}$

Passage-74

A point source is emitting light of wavelength 6000 $\overset{0}{A}$ is placed at a very small height h above a flat reflecting surface MN as shown in the figure. The intensity of the reflected light is 36% of the incident intensity. Interference fringes are observed on a screen placed parallel to the reflecting surface at a very large distance D from it



220. The shape of the interference fringes on the screen is

*(A) circle (B) ellipse (C) parabola (D) straight line

221. If the intensity at p be maximum then the minimum distance through which the reflecting surface MN should be displaced so that at P again becomes maximum?

*(A) 3×10^{-10} m (B) 6×10^{-10} m (C) 1.5×10^{-10} m (D) 12×10^{-10} m

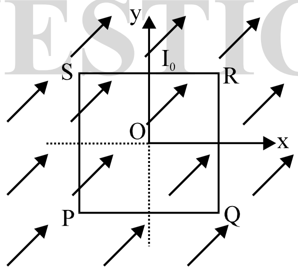
222. Ratio of maximum to minimum intensities at P

(A) 2 : 1 (B) 4 : 1 (C) 8 : 1 *(D) 16 : 1

MAGNETISM

Passage -75

A uniform; constant magnetic field $\vec{B}$ is directed at an angle of 45° to the x axis in x-y plane. PQRS is a rigid square wire frame carrying a steady current I_0, with its centre at the origin O. At time t = 0, the frame is at rest in the position shown in the figure, with its side parallel to x and y axis. Each side of the frame is of mass M and length L .



223. What is the torque $\vec{\tau}$ acting on the frame due to magnetic field?

(A) $\frac{I_0L^2B}{\sqrt{2}}(\hat{i}+\hat{j})$ (*B) $\frac{I_0L^2B}{\sqrt{2}}(-\hat{i}+\hat{j})$ (C) $\frac{I_0L^2B}{2}(-\hat{i}-\hat{j})$ (D) $\frac{I_0L^2B}{2}(\hat{i}+\hat{j})$

224. Find the angle by which the frame rotates under the action of this torque in a short interval of time Δt. (Δt is so short that any variation in the torque during this interval may be neglected.)

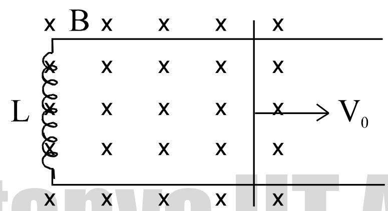
(*A) $\frac{3I_0B}{4M}(\Delta t)^2$ (B) $\frac{4I_0B}{M}(\Delta t)^2$ (C) $\frac{2I_0B}{3M}(\Delta t)^2$ (D) $\frac{4I_0B}{3M}(\Delta t)^2$

225. About which axis system will tend to rotate?

(*A) SQ (B) PQ (C) PS (D) PR

Passage-76

A coil of inductance 'L', connects the upper ends of two horizontal rails having zero resistance . A horizontal conductor of mass m is projected with initial velocity v_0 at $t = 0$, all the time maintaining contact with the two horizontal rails as shown . A uniform magnetic field of magnitude B exists in the region normal to the plain of the rails . The distance between the horizontal rails is l .



226. During the motion of conductor :
(A) Its velocity continuously decreases
(B) It will come to rest after some time
(*C) It will come to rest after some time and then returns in opposite direction
(D) Its velocity remains constant

227. The motion of conductor is :
(A) Non uniform with exponentially decreasing velocity
(*B) Non uniform with sinusoidally varying velocity
(C) Uniform
(D) None of these

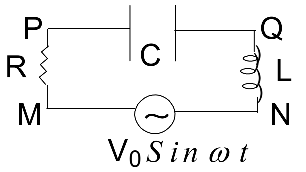
228. Magnetic energy in the inductor :

(A) is maximum at t=0 (B) is maximum at t= $\frac{2\pi\sqrt{m}}{Bl}$

(C) is maximum at t = $\frac{\pi\sqrt{m}}{Bl}$ (*D) none of these

EMI & AC
Passage-77

There is a series LCR circuit (as shown). An alternating source of emf having voltage $V = V_0 \text{ Sin } \omega t$ is applied between M & N.



Here $V_M - V_N = V_0 \; Sin\omega t$ and $\frac{1}{\omega C} - \omega L = R$.

229. The potential difference across R has got a phase difference of θ from that of alternating source of emf. where

(A) $\theta = \frac{\pi}{2}$ (B) $\theta = 0$ *(C) $\theta = \frac{\pi}{4}$ (D) $\theta = \pi$

230. R.M.S value of potential difference across capacitor will be

(A) $\frac{V_0}{\sqrt{2}}$ (B) $V_0 R \;\omega C$ (C) Zero *(D) $\frac{V_0}{2R\;\omega\;C}$

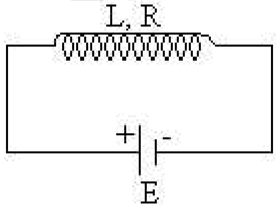
231. Potential difference across inductor i.e. $V_Q - V_N$ is

*(A) $\frac{V_0}{\sqrt{2}R}\omega L \,\text{Sin}\left(\omega t + \frac{3\pi}{4}\right)$ (B) $\frac{V_0 \omega L}{\sqrt{2}R}\text{Sin}\,\omega t$

(C) $\frac{V_0 \omega L}{\sqrt{2}R}\text{Sin}\left(\omega t + \frac{\pi}{2}\right)$ (D) $\frac{V_0 \omega L}{\sqrt{2}R}\text{Cos}\,\omega t$

Passage-78

A solenoid of resistance R and inductance L has a piece of soft iron inside it. A battery of emf E and of negligible internal resistance is connected across the solenoid as shown in the figure. At an instant, the piece of soft iron is pulled out suddenly so that inductance of the solenoid decreased to ηL with battery remain connected



232. The work done to pull out the soft iron piece

(A) $\frac{\eta LE^2}{2R^2}$ (B) $\frac{(1-\eta)LE^2}{2R^2}$ (C) $\frac{(1-\eta)LE^2}{\eta R^2}$ *(D) $\frac{(1-\eta)LE^2}{2\eta R^2}$

233. Assuming $t=0$ is the instant when iron piece has been pulled out, the current as a function of time is

*(A) $i = \frac{E}{R}\left[1-\left(1-\frac{1}{\eta}\right)e^{-t/\lambda}\right]$ (B) $i = \frac{E}{R}\left[1+\left(1+\frac{1}{\eta}\right)e^{-t/\lambda}\right]$

(C) $i = \frac{E}{R}\left[1-\left(1+\frac{1}{\eta}\right)e^{-t/\lambda}\right]$ (D) $i = \frac{E}{R}\left[1+\left(1-\frac{1}{\eta}\right)e^{-t/\lambda}\right]$ (where $\lambda = \frac{\eta L}{R}$)

234. Power supplied by teh battery as a function of time

*(A) $P = \frac{E^2}{R}\left[1-\left(1-\frac{1}{\eta}\right)e^{-t/\lambda}\right]$ (B) $P = \frac{E^2}{R}\left[1+\left(1+\frac{1}{\eta}\right)e^{-t/\lambda}\right]$

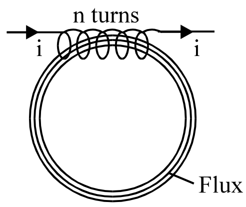
(C) $P = \frac{E^2}{R}\left[1-\left(1+\frac{1}{\eta}\right)e^{-t/\lambda}\right]$ (D) $P = \frac{E^2}{R}\left[1+\left(1-\frac{1}{\eta}\right)e^{-t/\lambda}\right]$

Passage-79

Magnetic circuit is defined as the route or path which is followed by magnetic flux. The laws of magnetic circuit are quite similar to (but not the same as) those of the electric circuit.

Consider an endless solenoid called a toroidal ring made of iron having a magnetic path of l-metre, area of cross-section A m^2 and a coil of N turns carrying I amperes, wound anywhere on it as shown in figure. Field strength inside the solenoid is

$$H = \frac{NI}{\ell}\text{Amp. Turn / m}$$



$B = \frac{\mu NI}{\ell}$ Amp. Turn / m

Total flux produced is

$$\phi = B \times A = \frac{\mu ANI}{\ell} \qquad \phi = \frac{NI}{\left(\frac{\ell}{\mu A}\right)} = \frac{NI}{s}$$

The numerator (NI) which produces magnetisation in the magnetic circuit is known as magnetomotive force (mmf).

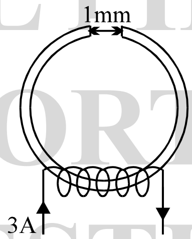
The denominator $\left(\frac{\ell}{\mu A}\right)$ is called the reluctance of the circuit and is analogous to resistance in electric circuit.

$\therefore$ flux $= \frac{\text{mmf}}{\text{reluctance}}$

Sometimes the above equation is called the 'Ohm's law of magnetic circuit' because it resembles a similar expression in electric circuit,

i.e., current $= \frac{\text{mmf}}{\text{resistance}}$

235. A steel ring of 15 cm mean radius and of circular section 1 cm has an air gap of 1 mm length. It is wound uniformly with 'n' turns of wire carrying a current of 3A. The air gap takes 60% of the total mmf which is 900 Amp.Turn. What is the value of 'n'?



(A) 900 turns (B) 700 turns *(C) 500 turns (D) 300 turns

236. A flux density of 1.2 Wb/m^2 is required in the 2 mm air gap of an electromagnet. Calculate the mmf required.

*(A) 1910 AT (B) 910 AT (C) 1500 AT (D) 1200 AT

237. A magnetic circuit consists of an iron ring of volume 960 cm^3 and mean circumference 80 cm. A current of 2 A in the magnetising coil of 200 turns produces a total flux of 1.2 mWb in the iron. The flux density in the ring is

*(A) 1 Wb/m^2 (B) 0.1 Wb/m^2 (C) 2 Wb/m^2 (D) 0.2 Wb/m^2

Passage-80

A coil having a resistance of 50.0Ω and an inductance of 0.5 H is connected to an AC source of 110 V, 50 Hz.

238. The impedance of the coil is (approximately)

*(A) 165Ω (B) 55Ω
(C) 330Ω (D) none of these

239. The rms current in the circuit is

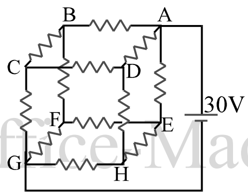
(A) 1.5 A *(B) 0.67 A
(C) 1 A (D) 3 A

240. If the frequency of the AC is decreased by a factor of 2, then the rms current

*(A) increases
(B) decreases
(C) remains uncharged
(D) may increase or decrease depending on the actual phase of the applied voltage

Passage-81

A resistor circuit is constructed such that twelve resistors are arranged to form a cube as shown in figure . Each resistor has a resistance of 2 ohm. The potential difference of 30 V is applied across two of the opposing points as shown.



241. The points having the same potential are :

(a) B, D, E (b) C, F, H (c) C, E
(A) only (a) is correct (B) (a), (b) and (c) are correct
(C) only (b) is correct *(D) (a) and (b) both are correct

242. If we replace resistors between A and B and resistors between G and H by resistors with wires of zero resistance, then the points having the same potential are :

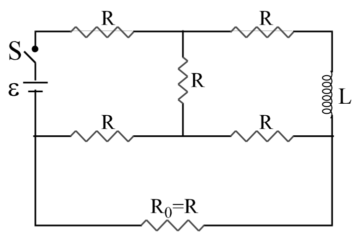
(a) D, E, C, F (b) A, B (c) G, H
(A) only (a) is correct (B) only (b) is correct
(C) only (c) is correct *(D) (a), (b), (c) are correct

243. In the above question, the potential difference between the points C and G is :

*(A) 15 V (B) 10 V (C) 20 V (D) 7.5 V

Passage-82

Consider a circuit consists of resistors, inductor, battery and a switch as shown. Resistance of resistors, inductance of inductor and EMF of battery are indicated. The switch is closed at t = 0.



244. Find current through inductor at steady state

(A) $\frac{\varepsilon}{2R}$ *(B) $\frac{\varepsilon}{4R}$ (C) $\frac{\varepsilon}{3R}$ (D) $\frac{\varepsilon}{8R}$

245. Find time constant of LR circuit during growth of current

(A) $\frac{L}{3R}$ *(B) $\frac{L}{2R}$ (C) $\frac{L}{R}$ (D) $\frac{2L}{R}$

246. After a long time switch is opened. Find heat generated through the resistor R_0

(A) $\frac{L\varepsilon^2}{32R^2}$ (B) $\frac{L\varepsilon^2}{320R^2}$ *(C) $\frac{L\varepsilon^2}{768R^2}$ (D) $\frac{L\varepsilon^2}{64R^2}$

MODERN PHYSICS

Passage-83

In a nuclear fusion reactor, the reaction occur in two stages :

(A) Two deuterium ($_1D^2$) nuclei fuse to form a tritium ($_1T^3$) nucleus with a proton as a biproduct. The reaction may be represented as D(D,p)T.

(B) A tritium nucleus fuses with another deutrium nucleus to form a helium ($_2He^4$) nucleus with a neutron as a biproduct. The reaction is represented as T(D, n) α.

Given : $m(_1D^2) = 2.014102$ u(atom)
$m(_1T^3) = 3.016049$ u(atom)
$m(_2He^4) = 4.002603$ u(atom)
$m(_1H^1) = 1.007825$ u(atom)
$m(_0n^2) = 1.008665$ u(atom)

247. The energy released in 1st stage of fusion reaction, is :
*(A) 4.033 MeV (B) 17.587 MeV
(C) 40.33 MeV (D) 1.7587 MeV

248. The energy released in the 2nd stage of fusion reaction :
(A) 4.033 MeV *(B) 17.587 MeV
(C) 40.33 MeV (D) 1.7587 MeV

249. What percentage of the mass energy of the initial deuterium is released ?
(A) 0.184 % (B) 0.284 %
*(C) 0.384 % (D) 0.484 %

Passage-84

A uniform monochromatic beam of light of wavelength 365×10^{-9} m and intensity 10^{-8} W/m^2, falls on a surface having absorption coefficient 0.8 and work function 1.6 eV.

250. The rate of number of electrons emitted per m^2, is given by :
*(A) 18.35×10^9 (B) 18.35×10^{10}
(C) 18.35×10^{11} (D) 18.35×10^{12}

251. Power absorbed per m^2 is given by :
(A) 6×10^{-9} W/m^2 (B) 7×10^{-8} W/m^2
*(C) 8×10^{-9} W/m^2 (D) 9×10^{-9} W/m^2

252. The maximum kinetic energy of the emitted photoelectrons is given by :
*(A) 1.8 eV (B) 2.8 eV
(C) 3.8 eV (D) 4.8 eV

Passage-85

"Whenever a radioactive element X undergoes β^- decay, daughter nucleus Y is formed in excited state , which on transition to normal state releases γ – radiation according to reaction .

$$X \longrightarrow Y^+ + \beta^- + \Delta E_1$$
$$Y^+ \longrightarrow Y + \gamma \text{ rad} + \Delta E_2$$

here M_X and M_Y are the atomic masses of elements X and Y respectively and m_e is the mass of β particle ."

253. The mass defect of the above decay is :
(A) $\Delta m^2 = M_X - M_Y - m_e$ (B) $\Delta m = M_X - M_Y - 2m_e$
*(C) $\Delta m = M_X - M_Y$ (D) none of these

254. In first step the energy released ΔE_1 is :

(A) $\Delta E_1 = \Delta mc^2$ *(B) $\Delta E_1 < \Delta mc^2$

(C) $\Delta E_1 > \Delta mc^2$ (D) none of these

255. In first step the energy released ΔE_1 is :

(A) In the form of kinetic energy of daughter nucleus Y

*(B) In the form of kinetic energy of Y and β particle

(C) In the form of heat and radiations

(D) None of these

Passage-86

"Whenever a radioactive element X undergoes β^- decay, daughter nucleus Y is formed in excited state , which on transition to normal state releases γ – radiation according to reaction .

$$X \longrightarrow Y^+ + \beta^- + \Delta E_1$$
$$Y^+ \longrightarrow Y + \gamma \text{ rad} + \Delta E_2$$

here M_X and M_Y are the atomic masses of elements X and Y respectively and m_e is the mass of β particle ."

256. The mass defect of the above decay is :

(A) $\Delta m^2 = M_X - M_Y - m_e$ (B) $\Delta m = M_X - M_Y - 2m_e$

(*C) $\Delta m = M_X - M_Y$ (D) none of these

257. In first step the energy released ΔE_1 is :

(A) $\Delta E_1 = \Delta mc^2$ (*B) $\Delta E_1 < \Delta mc^2$ (C) $\Delta E_1 > \Delta mc^2$ (D) none of these

258. In first step the energy released ΔE_1 is :

(A) In the form of kinetic energy of daughter nucleus Y

(*B) In the form of kinetic energy of Y and β particle

(C) In the form of heat and radiations

(D) None of these

Passage-87

In a photoelectric effect set-up a point source of light of power 3.2×10^{-3} W emits monoenergetic photons of energy 5 eV. The source is located at a distance of 0.8, from the centre of a stationary metallic sphere of work function 3 eV and of radius 8×10^{-3} m. The efficiency of photoelectron emission is one for every 10^6 incident photons. Assume that the sphere is isolated and initially neutral and that photoelectrons are instantly swept away after emission.

259. The number of photoelectrons emitted per second is

(*A) 10^5 per sec (B) 10^3 per sec (C) 10^7 sec (D) 10^6 sec

260. The ratio of the wavelength of incident light to the de-Brogile wavelength of the fastest photoelectrons emitted is

(*A) 285.1 (B) 255.3 (C) 265.3 (D) none

261. The time t is

(A) 105 sec (*B) 111 sec (C) 150 sec (D) none

Passage-88

A radio nuclide with decay constant λ is produced in a nuclear reactor at a rate q_0t per second, where q_0 is constant and t is time. During each decay energy E_0 is released. If at t = 0, production of radio nuclide started, find

262. Which differential equation correctly represent the above process?

*(A) $\frac{dN}{dt} + \lambda N = q_0 t$ (B) $\frac{dN}{dt} - \lambda N = q_0 t$ (C) $\frac{dN}{dt} + q_0 t = \lambda N$ (D) $\frac{dN}{dt} + q_0 t = -\lambda N$

263. Instantaneous power developed at any time t :

*(A) $\left(q_0t-\frac{q_0}{\lambda}+\frac{q_0}{\lambda}e^{-\lambda t}\right)E_0$ (B) $\left(q_0t+\frac{q_0}{\lambda}-\frac{q_0}{\lambda}e^{-\lambda t}\right)E_0$

(C) $\left(q_0t+\frac{q_0}{\lambda}+\frac{q_0}{\lambda}e^{-\lambda t}\right)E_0$ (D) $\left(q_0t-\frac{q_0}{\lambda}-\frac{q_0}{\lambda}e^{-\lambda t}\right)E_0$

264. Average power developed in any time t :

*(A) $\left(\frac{q_0t}{2}-\frac{q_0}{\lambda}+\frac{q_0}{\lambda^2t}-\frac{q_0}{\lambda^2t}e^{-\lambda t}\right)E_0$ (B) $\left(\frac{q_0t}{2}+\frac{q_0}{\lambda}+\frac{q_0}{\lambda^2t}-\frac{q_0}{\lambda^2t}e^{-\lambda t}\right)E_0$

(C) $\left(\frac{q_0t}{2}-\frac{q_0}{\lambda}+\frac{q_0}{\lambda^2t}+\frac{q_0}{\lambda^2t}e^{-\lambda t}\right)E_0$ (D) $\left(\frac{q_0t}{2}+\frac{q_0}{\lambda}+\frac{q_0}{\lambda^2t}+\frac{q_0}{\lambda^2t}e^{-\lambda t}\right)E_0$

Passage-89

Two metallic plates A and B, each of area 5 x $10^{-4}m^2$, are placed parallel to each other at a separation of 1 cm. Plate B carries a positive charge of 33.7 x 10^{-12}C. A monochromatic beam of light, with photons of energy 5 eV each, starts falling on plate A at t = 0 so that 10^{16} photons fall on it per square meter per second. Assume that one photoelectron is emitted for every 10^6 incident photons. Also assume that all the emitted photoelectrons are collected by plate B and the work function of plate A remains constant at the value 2 eV.

265. The kinetic energy of the most energetic photo-electron emitted at t = 10s when it reaches the plate B is

(A) 3 eV (B) 20 eV *(C) 23 eV (D) 17 eV

266. No. of photoelectrons emitted up to 10 sec

*(A) 5 x 10^7 (B) 2 x 10^6 (C) 5 x 10^6 (D) 2 x 10^7

267. Magnitude of electric field between plates A and B at t = 10 sec

(A) 1000 V/m *(B) 2000 V/m (C) 1500 V/m (D) 2500 V/m

Passage-90

When radioactivity was discovered, only three kind of radioactive decays α, β and γ were known. In the later years two more kinds of radioactive decay were discovered. According to the Pauli in β -decay process along with emission of electron another particle are also emitted called neutrino and antinutrino. The mass and charge on both the particles are zero and spin of both are 1/2 in the unit of $\frac{h}{2\pi}$. Spin of neutrino is antiparallel to its momentum where as spin of antinutrino is parallel to its momentum. The neutrino hypothesis saves the principles of energy conservation and angular momentum conservation in β -decay.

268. In which equation X-represent β^+- decay

(A) ${}_6C^{14} \to {}_7N^{14}+X+\bar{v}$ *(B) ${}_{29}Cu^{64} \to {}_{28}Ni^{64}+X+v$

(C) ${}_{29}Cu^{64}+\beta^- \to {}_{28}Ni^{64}+X$ (D) ${}_{92}U^{238} \to {}_{90}Th^{234}+X$

269. Which of the following decay is accompanied by x-ray

(A) β -decay (B) position - emission *(C) Electron capture (D) Gamma decay

270. Choose the correct option

*(A) Electron energy of β -particle vary from zero to a maximum for a particular nuclide

(B) The direction of emitted electrons and the recoiling nuclei are exactly opposite.

(C) End point energy E max is more than the energy equivalent to the mass difference between parant and daughter nuclide.

(D) Neutrino hypothesis violets the principle of conservation of angular momentum.

Passage-91

We have two radioactive nuclei A and B. Both convert into a stable nucleus C. Nucleus A converts into C after emitting two α-particles and three β-particles. Nucleus B converts into C after emitting one α-particle and five β-particles. At time t = 0, nuclei of A are 4 N_0 and that of B are N_0. Half-life of A (into the conversion of C) is 1 minute and that of B is 2 minutes. Initially number of nuclei of C are zero.

271. If atomic numbers and mass numbers of A and B are Z_1, Z_2, A_1 and A_2 respectively. Then :

(A) $Z_1 - Z_2 = 6$ *(B) $A_1 - A_2 = 4$

(C) both (a) and (b) are correct (D) both (a) and (c) are wrong

272. What are number of nuclei of C, when number of nuclei of A and B are equal ?

(A) 2 N_0 (B) 3 N_0

*(C) $\frac{9N_0}{2}$ (D) $\frac{5N_0}{2}$

273. At what time rate of disintegrations of A and B are equal ?

(A) 4 minutes *(B) 6 minutes

(C) 8 minutes (D) 2 minutes

Passage-92

A shell of space statation is a blackened sphere in which a temperature T = 500 K is maintained due to the operation of appliances of the station. Now, the station is enveloped by a thin spherical layer of black screen of nearly the same radius as the radius of the shell. Assume the temperature of the outer space to be zero.

274. The temperature of the black screen

(A) < 500 K *(B) = 500 K (C) > 500 K (D) None of these

275. The new temperature of the station shell will be

(A) < 500 K (B) = 500 K *(C) > 500 K (D) None of these

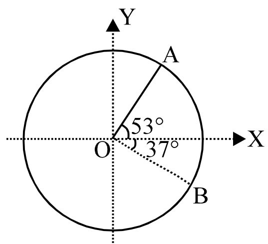
276. The net heat radiated by the station shell is

(A) less than earlier *(B) same as earlier (C) more than earlier (D) None of these

ROTATIONAL MOTION

Passage-93

A disc of radius 10 cm rotates in XY plane about an axis passing through its centre and perpendicular to the plane. At a particular moment a point A on the disc has an acceleration $\vec{a}_A = -6\hat{i}\ m/s^2$.



277. The angular speed of disc is :

(A) 6 rad/s clockwise (B) 3 rad/s clockwise

*(C) 6 rad/s anticlockwise (D) 3 rad/s clockwise or anticlockwise

278. The angular acceleration of disc is :

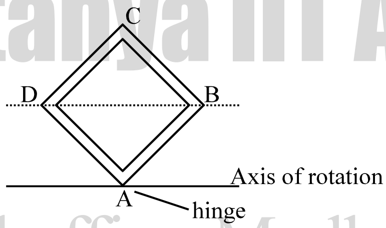
*(A) 48 rad/s^2 anticlockwise (B) 12 rad/s^2 anticlockwise
(C) 48 rad/s^2 clockwise (D) 12 rad/s^2 clockwise

279. The linear acceleration of point B on disc is :

(A) $6\hat{i}\ m/s^2$ *(B) $-6\hat{i}\ m/s^2$ (C) $6\hat{j}\ m/s^2$ (D) $-6\hat{j}\ m/s^2$

Passage-94

A square frame, made of 4 equal rods of length *l*, mass m each is hinged at corner A so that it can rotate about a horizontal axis parallel to line DB as shown. It is slightly disturbed and it starts falling about hinge axis. Calculate following parameters when the frame is completely inverted and point C reaches the lower point of its motion.



280. Angular velocity of frame :

*(A) $\sqrt{3\sqrt{2}\frac{g}{\ell}}$ (B) $\sqrt{\frac{3g}{\ell}}$ (C) $\sqrt{\frac{3}{\sqrt{2}}\frac{g}{\ell}}$ (D) $\sqrt{\frac{6g}{\ell}}$

281. Velocity of point B :

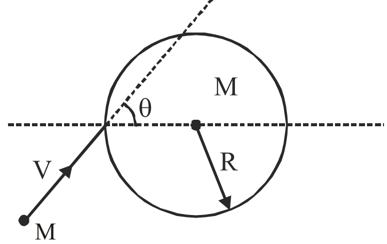
*(A) $\sqrt{\frac{3}{\sqrt{2}}g\ell}$ (B) $\sqrt{3\sqrt{2}\,g\ell}$ (C) $\sqrt{3\frac{g}{\ell}}$ (D) $\sqrt{6\frac{g}{\ell}}$

282. Hinge reaction :

(A) 4 mg (B) 8 mg *(C) 12 mg (D) 16 mg

Passage-95

If external forces on a system is zero, linear momentum remains conserved. Co–efficient of restitution 'e' is defined as ratio of velocity of separation and velocity of approach along line of impact. A disc of mass M and radius of curvature R is lying on the ground (frictionless). A small ball of mass M moving on the ground with a velocity V strikes the disc and collides with it. The coefficient of restitution being 'e' :



283. Velocity of approach for the system (ball + disc) is :

*(A) V cos θ (B) V sin θ
(C) V (D) zero

284. Velocity of the disc after collision is :

(A) $\frac{V\cos\theta}{2}$ (B) $\frac{eV\cos\theta}{2}$

*(C) $\frac{V\cos\theta(1+e)}{2}$ (D) $\frac{V\cos\theta(1-e)}{2}$

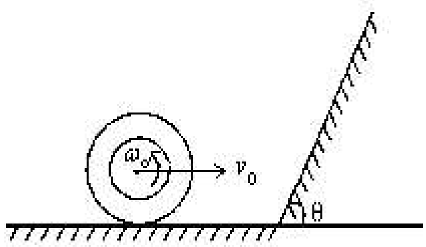
285. The speed of the ball after the collision :

(A) $eV\cos\theta$ *(B) $V\sqrt{\frac{\cos^2\theta(1-e)^2}{4}+\sin^2\theta}$

(C) $V\sqrt{\cos^2\theta(1+e)^2+\frac{\sin^2\theta}{4}}$ (D) $V\sqrt{\frac{\cos^2\theta(1+e)^2}{4}+\sin^2\theta}$

Passage-96

A ring of masss m and radius r is projected with velocity v_0 while spinning with angular velocity $\frac{v_0}{4r}$ at the bottom of an inclined plane of angle of inclination $\theta = 60^o$ as shown in fig. The ring begins to roll without slipping after colliding with the inclined plane. Before collision sense of rotation is depicted in the fig .



286. Velocity of center of mass of the ring just after the collision is :

(A) v_0 (B) $\frac{v_0}{2}$ (C) $\frac{v_0}{4}$ *(D) $\frac{v_0}{8}$

287. Loss of kinetic energy of ring during collision is :

(A) $\frac{9}{32}mV_0^2$ *(B) $\frac{33}{64}mV_0^2$ (C) $\frac{1}{2}mV_0^2$ (D) zero

288. If the ring after collision rolling up the incline then maximum height attained by it is :

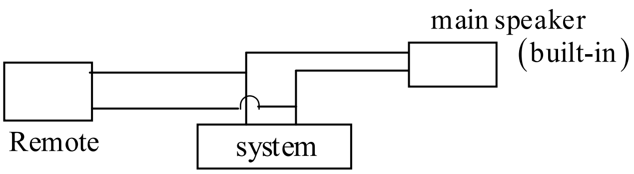
(A) $\frac{V_0^2}{2g}$ *(B) $\frac{V_0^2}{128g}$ (C) $\frac{V_0^2}{64g}$ (D) $\frac{V_0^2}{32g}$

CURRENT ELECTRICITY

Passage-97

Most of the times we connect remote speakers to play music in another room along with the buit-in speakers. These speakers are connected in parallel with the music system.

At the instant represented in the picture, the a.c. voltage across the speakers is 6.00 V. The main speaker resistance is $8.00\,\Omega$ and the remote speaker has $4.00\,\Omega$ resistance.



289. The equivalent resistance of the speakers is :

(A) 12Ω (B) 4Ω *(C) 2.67Ω (D) 0.375Ω

290. The total current supplied by the music system is :

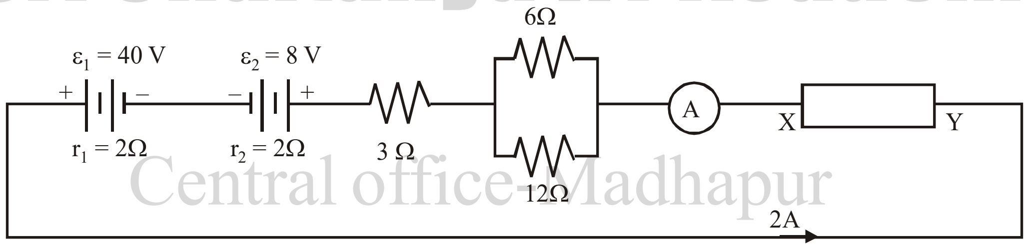
*(A) 2.25 A (B) 1 A (C) 16 A (D) 1.5 A

291. The power dissipated in the speaker of 4.00Ω resistance is :

*(A) 9 watts (B) 4.5 watts (C) 13.5 watts (D) 0

Passage-98

In the circuit shown the ammeter 'A' reads 2A. X Y is a circuit element which can either be a resistance or a battery. In the direction of the current there will always be a potential drop across the resistance. Regarding battery you have to remember that the potential difference across its terminals can be more than its emf or less depending upon whether the battery is delivering the current, or is being charged in the circuit



292. Assuming XY to be a resistor, its value is :

*(A) 5Ω (B) 9Ω (C) 13Ω (D) 9.75Ω

293. Assuming XY to be a battery with 2Ω internal resistance which is being charged, its emf is :

*(A) 6 V (B) -6 V (C) 0 (D) 2 volts

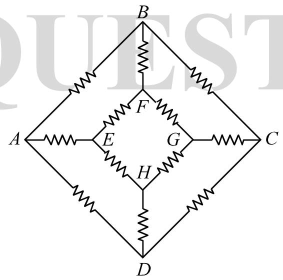
294. The potential difference as you move from Y to X while considering it to be a battery is :

*(A) - 10 V (B) 10 V (C) 0 (D) 2 volts

Passage-99

In the network shown in figure, each resistance is R :

295. The equivalent resistance between A and B is :



(A) $\frac{3R}{4}$ (B) $\frac{5R}{6}$ *(C) $\frac{7R}{12}$ (D) $\frac{4R}{3}$

296. The equivalent resistance between A and G is :

(A) $\frac{3R}{4}$ *(B) $\frac{5R}{6}$ (C) $\frac{7R}{12}$ (D) $\frac{4R}{3}$

297. The equivalent resistance between A and D is :

(A) $\frac{3R}{4}$ (B) $\frac{5R}{6}$ *(C) $\frac{7R}{12}$ (D) $\frac{4R}{3}$

Passage-100

Potentiometer is an ideal voltmeter as voltmeter draws some current through the circuit while potentiometer needs no current to work. Potentiometer works on the principle of e.m.f. comparison. In working condition, a constant current flows through out the wire of potentiometer using standard cell of e. m. f. e_1. The wire of potentiometer is made of uniform material and cross sectional area and it has uniform resistance per unit length. The potential gradient, depends upon the current in the wire. A potentiometer with a cell of e. m. f. 2 V and internal resistance 0.4 Ω is used across the wire AB. A standard cadmium cell of e. m. f. 1.02 V gives a balance point at 66.3 cm length of wire. The standard cell is then replaced by a cell of unknown e. m. f. e and the balance point found similarly turns out to be 82.3 cm length of the wire. The length of potentiometer wire AB is 1 m .

298. The value of e is :

*(A) 1.26 V (B) 2.63 V (C) 1.83 V (D) none

299. The reading of potentiometer if 4 V battery is used instead of e, is :

(A) 88.3 cm (B) 47.3 cm (C) 95 cm *(D) can not be calculated

300. If the resistance is connected across the cell e, the balancing length will be :

(A) increase (B) decrease *(C) remains same (D) None

ANSWER KEY - PHYSICS

Q.No.	Ans.	Q.No.	Ans.	Q.No.	Ans.	Q.No.	Ans.	Q.No.	Ans.	Q.No.	Ans.
1	B	51	D	101	B	151	A	201	A	251	C
2	C	52	B	102	D	152	C	202	A	252	A
3	C	53	B	103	A	153	B	203	B	253	C
4	A	54	B	104	D	154	A	204	A	254	B
5	B	55	A	105	C	155	B	205	D	255	B
6	D	56	B	106	A	156	D	206	A	256	C
7	B	57	C	107	B	157	B	207	A	257	B
8	C	58	A	108	C	158	B	208	D	258	B
9	B	59	B	109	D	159	A	209	C	259	A
10	B	60	C	110	C	160	B	210	A	260	A
11	A	61	D	111	C	161	C	211	B	261	B
12	D	62	B	112	D	162	D	212	C	262	A
13	B	63	D	113	D	163	A	213	D	263	A
14	C	64	A	114	B	164	C	214	B	264	A
15	A	65	C	115	B	165	A	215	A	265	C
16	B	66	B	116	B	166	A	216	D	266	A
17	A	67	A	117	C	167	B	217	D	267	B
18	B	68	B	118	D	168	A	218	A	268	B
19	C	69	C	119	A	169	A	219	B	269	C
20	C	70	B	120	D	170	A	220	A	270	A
21	D	71	A	121	B	171	D	221	A	271	B
22	A	72	D	122	B	172	B	222	D	272	C
23	C	73	C	123	C	173	D	223	B	273	B
24	D	74	B	124	A	174	C	224	A	274	B
25	A	75	A	125	C	175	D	225	A	275	C
26	C	76	C	126	B	176	A	226	C	276	B
27	C	77	B	127	B	177	B	227	B	277	C
28	C	78	A	128	A	178	B	228	D	278	A
29	C	79	B	129	C	179	B	229	C	279	B
30	B	80	C	130	A	180	B	230	D	280	A
31	D	81	A	131	C	181	A	231	A	281	A
32	A	82	B	132	B	182	C	232	D	282	C
33	C	83	C	133	C	183	C	233	A	283	A
34	C	84	B	134	D	184	B	234	A	284	C
35	B	85	A	135	C	185	C	235	C	285	B
36	A	86	D	136	A	186	D	236	A	286	D
37	B	87	B	137	B	187	A	237	A	287	B
38	C	88	B	138	A	188	A	238	A	288	B
39	D	89	A	139	A	189	A	239	B	289	C
40	B	90	D	140	D	190	A	240	A	290	A
41	A	91	C	141	C	191	C	241	D	291	A
42	C	92	B	142	A	192	B	242	D	292	A
43	B	93	A	143	B	193	A	243	A	293	A
44	B	94	A	144	B	194	B	244	B	294	A
45	B	95	A	145	D	195	A	245	B	295	C
46	D	96	D	146	C	196	A	246	C	296	B
47	A	97	C	147	B	197	C	247	A	297	C
48	B	98	A	148	C	198	B	248	B	298	A
49	C	99	A	149	A	199	D	249	C	299	D
50	B	100	C	150	A	200	A	250	A	300	C